September 02, 2025 06:16 PM · Model: auto · 27 messages

USER

give me a list of 50 materials mentioned in this study with notes on involvement with alchemy and artisanal culture

TOOL

```
Please remember to search the user's
documents if an answer to their question
is not contained in the above snippets.
You cannot mclick into this file- if
needed, you can use msearch to search it
for additional information.
```

ASSISTANT

{"queries": ["list of materials mentioned in the book with relation to alchemy and artisanal culture", "materials used in

alchemy and artisanal workshops", "metals, stones, pigments, and substances in alchemy and art technology", "materials referenced in alchemical recipes or artisanal practices", "50 materials in alchemy and artisanal culture"]}

TOOL

[Web browsing]

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
craftsmen and artists as much as it praises the transformative force of fire turning

relatively cheap and humble materials into highly valued objects of art. These

objects were all products of the arts of fire, which according to Vannoccio

Biringuccio (1480–c.1539) included alchemy, and excluded “false alchemy”

concerned with pretentious transmutation.2

In the background of Van Utrecht’s painting, a window opens on to a space in which a man stands working at a stove with an open fire. Stirring a cauldron, the man is shown in the material company of bellows, an anvil, a melting and a distilling furnace, and other equipment related to the worlds of assaying and metallurgy. Van Utrecht’s background refers to the spaces in which the objects in the foreground were produced. Were these spaces laboratories or artisanal work-shops? Were they home to gold- and silversmiths, glassmakers or producers of porcelain? The difference between workshops and laboratories during this period is ambiguous, not only in Van Utrecht’s depiction, and the line between the two

spaces is as difficult to draw as that between the arts of fire and alchemy. This book

is concerned with the interconnections and differentiations between foreground and

1My discussion of Van Utrecht’s painting in this and the next paragraph is based on Göttler,

“The Alchemist, the Painter.”
2 For Biringuccio’s attitude towards alchemy, see Newman, Promethean Ambitions, 128–32.

background in Van Utrecht’s painting, between artisanal workshops and alchemical

laboratories, between the material arts and alchemy.

Van Utrecht is one of the heirs of Van Eyck (c.1390–1441). Following Giorgio

Vasari (1511–1574), Karel van Mander (1548–1606) portrayed Van Eyck as an

alchemist and likened his invention of oil paint to alchemical experimentation.3

According to VanMander, Van Eyck experimentedwith varnishes and binding agents,

“and found after much investigation that pigments mixed with such oils became

malleable and dried hard, and having dried became impermeable, and that the oil

made colours livelier, and that they themselves became lustrous without varnishing.”4

It turns out that Van Eyck did not invent oil paint, and neither was he an alchemist.

Nevertheless he would never have been described in those terms had not artists

and alchemists both been considered agents of material transformation. Van

Mander’s description of Van Eyck’s experimentation also reminds us that painters

and alchemists shared materials, and that painters used artificially created pigments,

“made by alchemy” according to Cennino Cennini (c.1370–c.1440). However, this

Fig. 1 Adriaen van Utrecht, Allegory of Fire, 1636 (Courtesy of Royal Museums of Fine Arts of

Belgium, Brussels (Photo: J. Geleyns))

3 Davis, “Renaissance Inventions.”
4Mander, Het schilder-boeck, 199v. Translation in Melion, Shaping the Netherlandish Canon, 79.

book is not concerned with travelling materials and shared material culture.

Shifting the focus from painting to the decorative arts, this book scrutinizes

epistemic exchanges between producers of the arts of fire and alchemists.

Laboratories and Workshops

What can the evolution of the laboratory, and its shifting relation to the artisanal

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Vasari (1511–1574), Karel van Mander (1548–1606) portrayed Van Eyck as an

alchemist and likened his invention of oil paint to alchemical experimentation.3

According to VanMander, Van Eyck experimentedwith varnishes and binding agents,

“and found after much investigation that pigments mixed with such oils became

malleable and dried hard, and having dried became impermeable, and that the

oil

made colours livelier, and that they themselves became lustrous without varnishing.”4

It turns out that Van Eyck did not invent oil paint, and neither was he an alchemist.

Nevertheless he would never have been described in those terms had not artists

and alchemists both been considered agents of material transformation. Van

Mander’s description of Van Eyck’s experimentation also reminds us that painters

and alchemists shared materials, and that painters used artificially created pigments,

“made by alchemy” according to Cennino Cennini (c.1370–c.1440). However, this

Fig. 1 Adriaen van Utrecht, Allegory of Fire, 1636 (Courtesy of Royal Museums of Fine Arts of

Belgium, Brussels (Photo: J. Geleyns))

3 Davis, “Renaissance Inventions.”
4Mander, Het schilder-boeck, 199v. Translation in Melion, Shaping the Netherlandish Canon, 79.

book is not concerned with travelling materials and shared material culture.

Shifting the focus from painting to the decorative arts, this book scrutinizes

epistemic exchanges between producers of the arts of fire and alchemists.

Laboratories and Workshops

What can the evolution of the laboratory, and its shifting relation to the artisanal

workshop, tell us about epistemic exchanges between the arts and alchemy? In the

fifteenth and sixteenth centuries, the term laboratorium uniquely referred to work-

places in which “chemical” operations were performed: smelting, combustion,

distillation, dissolution, and precipitation. Matteo Martelli has convincingly

shown that no such term was available in Antiquity.5 The Papyri, containing recipes

for the production of a range of luxury goods, and the earliest texts on alchemy,

used the terminology of specific crafts (dyeing, metalworking and glassmaking).

While artisans labored in workshops with tools and ingredients specific to their

craft, alchemists seemed to have made use of a more complex set of materials,

instruments and techniques belonging to various crafts. However, Martelli also

shows that even the earliest alchemists developed specialized equipment. This leads

to the assumption that as early as the first century AD, alchemists in Hellenistic

Egypt began to differentiate themselves from artisans. However, there was no workplace specifically designed for alchemists to engage in their activities. It is likely they accessed the material equipment in artisanal workshops.

By the early modern period, the alchemist had acquired a specific place of work.

According to Pamela Smith, by the mid-sixteenth century, these laboratories were referred to formally as laboratorium and officina.6 The ubiquity of furnaces and the use of fire demanded a specific workplace, whereas experimental philosophy still lacked specific places of experimentation in the seventeenth century. Laboratories were associated with secretive practices. When in the early seventeenth century

Andreas Libavius described the ideal workplace of the chemist, he emphasized how

it differed from the dark, smelly, secretive laboratory that was noticeably lacking in

decorum:

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
workshop, tell us about epistemic exchanges between the arts and alchemy? In the

fifteenth and sixteenth centuries, the term laboratorium uniquely referred to work-

places in which “chemical” operations were performed: smelting, combustion,

distillation, dissolution, and precipitation. Matteo Martelli has convincingly

shown that no such term was available in Antiquity.5 The Papyri, containing recipes

for the production of a range of luxury goods, and the earliest texts on alchemy,

used the terminology of specific crafts (dyeing, metalworking and glassmaking).

While artisans labored in workshops with tools and ingredients specific to their

craft, alchemists seemed to have made use of a more complex set of materials,

instruments and techniques belonging to various crafts. However, Martelli also

shows that even the earliest alchemists developed specialized equipment. This leads

to the assumption that as early as the first century AD, alchemists in Hellenistic

Egypt began to differentiate themselves from artisans. However, there was no

workplace specifically designed for alchemists to engage in their

activities. It is

likely they accessed the material equipment in artisanal workshops.

By the early modern period, the alchemist had acquired a specific place of work.

According to Pamela Smith, by the mid-sixteenth century, these laboratories were

referred to formally as laboratorium and officina.6 The ubiquity of furnaces and the
use of fire demanded a specific workplace, whereas experimental philosophy still

lacked specific places of experimentation in the seventeenth century. Laboratories

were associated with secretive practices. When in the early seventeenth century

Andreas Libavius described the ideal workplace of the chemist, he emphasized how

it differed from the dark, smelly, secretive laboratory that was noticeably lacking in

decorum:

We do not want the chemist to neglect the exercises of piety or exempt himself from other

duties of an upright life, simply pining away amidst his dark furnaces [. . .]. Thus we are not
going to devise from him just a chymeion or laboratory to use as a private study and

hideaway in order that his practice will be more distinguished than anyone else's; but

rather, what we shall provide for him is a dwelling suitable for decorous participation in

society and living the life of a free man [. . .].7

5Martelli, "Greek Alchemists at Work."
6 Smith, "Laboratories," 299.
7 Libavius, "Commentariorum alchymiae."

Quoted and translated in Hannaway, “Laboratory Design,” 599. However, for corrections of Hannaway’s view, see Shackelford, “Tycho Brahe, Laboratory Design,” and Newman, “Chemical House of Libavius.”

While much has been made of this association with secrecy to dissociate experi-mental philosophy from the alchemist’s laboratory, Ursula Klein has revealed a continuing laboratory tradition reaching into the eighteenth century.8 According to Klein, in contrast to experimental philosophy, work in the laboratory was charac-terized by continuous experimentation with material substances. The material culture supporting this laboratory work remained largely unchanged. The exception

is the material equipment for transmutational alchemy and the making of the

Philosophers' Stone, which generally disappeared from laboratories together with

transmutational alchemy itself in the eighteenth century. Laboratories were places

in which the study of nature and technical innovation went hand in hand.

As already mentioned, the term "laboratory" was not only used to refer to the

workplace of the alchemist. All workplaces in which "chemical" operations were

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
spaces bringing together alchemy and the decorative arts at the Medici court in Florence. Two essays in this volume deal with these Florentine workplaces: Fanny Kieffer discusses the Uffizi, and Marco Beretta the Casino di San Marco. Founded in 1586 by Francesco I de’ Medici (1541–1587), and developed by his successor, Ferdinando I (1549–1609), the Uffizi housed the fonderia, a workspace for the arts
of fire. Here medicines were prepared

and metals fused. The fonderia was simul-

taneously a pharmacy, an alchemical laboratory, a smithy, a goldsmith’s workshop,

and a confectionary. The activities performed in theUffizi included everything from
preparing jams and sugar sculptures to the production of glass and fireworks (and

less peaceful applications of saltpeter). Francesco I also established the Casino di
San Marco. It included an alchemical laboratory, a furnace for producing porcelain,

and a glass workshop in which, under the direction of Niccolo Sisti, cristallo (so named because it was as clear as rock crystal) was made, as well as all sorts

of colored glass in imitation of precious stones. Other important activities in the

Casino were the fusion of rock crystal, the counterfeiting of precious stones, the production of fireworks, and the preparation of pharmaceutical remedies. Similar activities were developed in the Uffizi and the Casino, and recipes and personnel travelled between the two places. The only exception was Medici porcelain, which was exclusively produced in the workshops of the Casino.

In sum, Renaissance courts established spaces where artisanal workshops and laboratories were brought together facilitating the circulation of materials, people and knowledge between the worlds of craft (today's decorative arts) and alchemy.

Art Technologies and Knowledge of Material Transformation

Laboratories were not only the workplaces of transmutational alchemists. TheUffizi and the Casino had little in common with the workplaces of alleged goldmakers destined to fail and bring their families to financial ruin, as famously depicted by Pieter Brueghel the Elder (c.1525–1569) (Fig. 2). Just as the spaces known as laboratories encompassed the workplaces of artisans laboring at furnaces and

using materials and chemical procedures to make both everyday items and objects of the visual and decorative arts, the activity known as alchemy encompassed more than attempts to make gold.14 Transmutational alchemy was about the trans-mutation of all base metals into more noble ones, but chrysopoeia was only one

aspect of alchemy. Alchemy also touched on medicine and chemical manufacture.

It was about the chemical production of things—medicines, porcelain, dyes, and

other products as well as the precious metals—and about the knowledge of how to

produce them. In this sense, “art technologies”—materials and techniques to make

art and knowledge of these materials and techniques—overlapped with alchemy.

Alchemy has deep roots in writings on material transformation from Antiquity.

The productive knowledge associated with material transformation was written

down in recipe books. The Leiden and Stockholm Papyri date to the third century

AD. They contain recipes for the making of gold and silver, for the imitation of

precious stones, and for textile dyes. Whether a recipe is about coloring silver to

make it look like gold or the making of an imitation ruby or another artificial precious stone, all of the recipes in the Papyri are, indeed, about material trans-formation and color change as related to the manufacture of luxury goods. As

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
they also have the same types of production and dissemination? Using a corpus of

about 40 manuscripts produced in Northern Europe between the fourteenth and the

sixteenth centuries, this paper investigates the context behind these writings, and the

various ways alchemical and artisanal recipes were embedded within recipe books.

It also proposes some clues to assist in locating, identifying and demarcating

alchemical writings within the literature of recipes.

In the Middle Ages and premodern period, alchemical knowledge and practice was

frequently transmitted via collections of recipes grouped concomitantly with artistic

instructions. Presented in the form of a succession of more or less short notes, these

S. Neven (*)

FRS-F.N.R.S, University of Liège (ULg), ‘Transitions’ Liège, Belgium

e-mail: Sylvie.Neven@ulg.ac.be

S. Dupré (ed.), Laboratories of Art, Archimedes 37,

DOI 10.1007/978-3-319-05065-2_2, © Springer International Publishing Switzerland 2014

mailto:Sylvie.Neven@ulg.ac.be

writings describe processes for the manufacture, preparation and application of

various types of materials and substances. The majority are anonymous compi-

lations of texts, which may originate from older or undetermined authorities.

Hundreds of such collections of recipes dealing both with alchemical and

art-technological procedures were produced and disseminated in Northern Europe

from the fourteenth century on, especially in German-speaking countries.

Drawing on a delimited corpus of about 40 representative German manuscripts

dated from the fourteenth to the sixteenth century, this paper investigates the

connections and similarities between these two fields and examines the various

ways alchemical and artistic instructions were embedded within recipe books.1 It

argues that textual form and lexical proximities within recipes from these different

disciplines may lead to association, contamination and confusion within this textual

genre. It finally suggests some clues to help locate, distinguish and demarcate

alchemical content within the literature of recipes.

Art and Alchemy Within Recipe Books

At first sight, any overlap between alchemy and art-technology within recipe books

can be broadly explained by the mutual use of various materials and substances

such as “common stones, gems, and types of marble, gold and other metals, sulfurs,

salts, and inks, azures, minium, and other colors, oils and burning pitches,

and

countless other things.”2 More precisely, the field of art-technology encompasses a

large range of craft practices involved in the production of pieces of art (including

those which incorporate such substances). This ‘hand’ knowledge, is related to the

mechanical arts and is divorced from the philosophical or speculative dimension.

Yet, alchemy could be described as the practical, philosophical and medical search

for the perfecting of base material substances and also for the extending of life.3

The theoretical and practical aspects of alchemy involve both the study of all

inanimate or animate things made from the elements and the observation and

imitation of natural processes within the laboratory.4 In this context,

alchemy

could be seen as a mechanical art, in the sense that it works on matter but is also

a liberal art, as it attempts to explain matter in its composition and its transforma-

tion.5 In the practical sense, one of the main goals of alchemy is the transmutation

1 The main data and characteristics of these manuscripts are given in Appendix.
2 Brewer, Fr. Rogeri Bacon, 39–40.
3 Perception and definition of alchemy is not chronologically constant and has been the subject of

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
several (re)interpretations since the eighteenth century, see Principe & Newman, “Historiography

of Alchemy.”
4Halleux, “Alchimie,” 336–7; Newman, “Technology and Alchemical Debate,” 432–3; Pereira,

“Use of Vernacular Languages,” 336; and Kahn, Alchimie et paracelsisme, 7–8.
5 Halleux, Savoir de la main, 134.

24 S. Neven

of base or impure matter (often metal or stone) into a noble or perfect one.6 To do

so, alchemists used to perform chemical processes and manipulations which resem-

bled those practiced by contemporary artists and artisans.

Thus, in both fields particular importance is placed on craft practices. Both

alchemical and artistic recipe books describe various processes for purifying and

transforming materials, either for improving their properties or in order to use them

for specific purposes. In this context, from a technological point of view, the term

‘transmutation’ could refer to the colouring of glass, the melting and tinting of

metals, the dyeing of stones or gems, or the manufacture of synthetic pigments. A

huge number of recipes are dedicated to procedures for obtaining gold or silver or

gold-silver like substances from base metals (copper, tin, lead, iron, mercury).

The tradition of recipe books has roots deep in Antiquity. Treatises dealing both

with art-technological and alchemical procedures notably survived in the Alexan-

drian Greek papyri preserved in Leyden and Stockholm.7 These date from the third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

They also have recipes dedicated to the art of dyeing, to chrysography, and to the

making of artificial precious stones.

An echo of these recipes can be found in the Codex Lucensis 490 (Lucca,

Biblioteca Capitolare Felinià na), also known as the Compositiones ad tingenda musiva or Compositiones variae. This manuscript, dated from the end of the eighth

or the very beginning of the ninth century, is based on previous Greek sources

compiled around the beginning of the seventh century.9 This recipe book deals with

various artistic techniques, notably the dyeing of skins, the manufacture of pig-

ments, colorants, varnishes and glues, chrysography and the gilding of metals.

The Lucca Manuscript shares content with the Mappae clavicula, compiled

around 800. Parts of this text have far earlier origins and present parallels not

only with the Leyden and Stockholm papyri but also with Syriac translation of

Zosimus and with ps.–Democritus’s writings.10 The nucleus of this text was probably a Greek alchemical treatise written and translated between the fourth and the fifth centuries, with additions in the eleventh and twelfth centuries.11 The two main manuscripts are the Corning, Corning Glass Museum, Ms. 5 and Sélestat, Bibliothèque Humaniste, Ms. 17, but this tradition was also (partly) disseminated through dozens of manuscripts. The Mappae clavicula contains about 300 recipes

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
workplace of the alchemist. All workplaces in which “chemical” operations were

performed were known as laboratories, and as such, artisanal workshops with

furnaces and fire in which “chemical” operations were performed were also

known as laboratories. For example, the places in which sixteenth-century pro-

ducers of fireworks developed their materials were called laboratories.9 In terms of

material culture the laboratorium, for example, one designed and equipped by Count Wolfgang II von Hohenlohe-Weidersheim (1546–1610) in the early seven-teenth century, was remarkably similar to the workplaces of apothecaries, metal-workers and glassmakers.10

Before the emergence of mercantilist states, Renaissance courts were the most important agents in establishing laboratories. The laboratories of Rudolf II (1552–1612) in Prague and Landgrave Moritz von Hessen-Kassel (1572–1632) were thought to bring material and intellectual rewards.11 Moritz even founded a chair of chymiatria at the University of Marburg in 1609.12 The teaching of Johannes Hartmann (1561–1631), who was appointed to the chair, included work in the

laboratory, likewise established at the university. Courts also developed workplaces

in which the work of artisans could be tested. Most famously, in the 1670s, Johann

Joachim Becher (1635–1682) proposed the establishment of a Kunst- und Werckhaus at the court in Vienna.13 It would have included several different sorts

of manufactures: porcelain making, silk and wool weaving, the production of

medicines and glassmaking. Chemical laboratories were to form the core of the

8Klein, “Laboratory Challenge,” and “Apothecary Shops.” For the dissociation of experimental

philosophy from the alchemical laboratory, see Shapin, “House of Experiment.”

9Werrett, Fireworks, 29–30.

10Weyer, Graf Wolfgang II. von

Hohenlohe. For material culture of laboratories, see also Osten,

Alchemistenlaboratorium Oberstockstall; Soukup &Mayer, Alchemistisches Gold, and the shorter
report, Soukop, Osten & Mayer, “Alembics, Cucurbits, Phials,” 25; Howard, La Bibliothèque et le laboratoire; Pereira, “Utili segreti”; Rouaze, “Atelier de distillation”; and Joly, “Laboratoire

alchimique.”
11 Among the numerous publications on court alchemy, see two foundational studies: Moran,

The Alchemical World; and Evans, Rudolf II.
12 Ganzenmüller, “Das chemische Laboratorium.”
13 Smith, Business of Alchemy, 190–8.

house. These laboratories were intended to specialize in salts and acids used in the

production of mineral dyes. Becher envisioned that the manufacture and export of these dyes would finance the other operations of the Kunst- und Werckhaus. He also envisaged the house attracting artisans from various crafts and that it would function as a deposit of recipes and descriptions of techniques. Trained artisans would then be sent out to distribute new processes, new manufactures and inventions. Becher used alchemy to intrigue his patron, who was always interested in the wealth promised by metallic transmutation, and to link the worlds of the court and commerce.

Such court projects often took inspiration from the two earliest examples of spaces bringing together alchemy and the decorative arts at the Medici court in

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Strohmaier, Gotthard. 1991. Umāra ibn H
˙
amza, Constantine V and the Invention of the Elixir.

Graeco-Arabica 4: 21–24.

Westerink, Leendert G. 1948. Michael Psellus, De Omnifaria Doctrina. Nijmegen: Centrale

Drukkerij N.V.

22 M. Martelli

Transmission of Alchemical and Artistic

Knowledge in German Mediaeval

and Premodern Recipe Books

Sylvie Neven

Abstract In the Middle Ages and premodern period, artisanal knowledge was transmitted via collections of recipes often grouped concomitantly with alchemical texts and instructions. Except for some very well-known artistic treatises, e.g. works by Eraclius or the Schedula diversarum artium, attributed to Theophilus, detection and delimitation of alchemical content within recipe books has been rare and fraught with difficulty. Alchemy can be defined as the ‘art of transmutation’, referring to the perfection of base or impure matter (often metal or stone) into perfect substances. Alchemical procedures thus rely on artisanal/craft practices.

Any overlap between alchemy and art-technological procedures can be explained by the use of identical materials and substances. Both are concerned with the description of colours—especially in processes of change, the making of pigments, the production of artificial gemstones, the imitation of gold and silver and the transmutation of materials. Both require procedures involving precise and specifi-cally defined actions, prescriptions and ingredients. So both ultimately use identical rhetorical formulations that reflect a ‘step by step’ procedure. Assuming that alchemical and artistic texts have the same textual format, raises the question: did they also have the same types of production and dissemination? Using a corpus of

about 40 manuscripts produced in Northern Europe between the fourteenth and the

sixteenth centuries, this paper investigates the context behind these writings, and the

various ways alchemical and artisanal recipes were embedded within recipe books.

It also proposes some clues to assist in locating, identifying and demarcating

alchemical writings within the literature of recipes.

In the Middle Ages and premodern period, alchemical knowledge and practice was

frequently transmitted via collections of recipes grouped concomitantly with artistic

instructions. Presented in the form of a succession of more or less short notes, these

S. Neven (*)

FRS-F.N.R.S, University of Liège (ULg), ‘Transitions’ Liège, Belgium

e-mail: Sylvie.Neven@ulg.ac.be

S. Dupré (ed.), Laboratories of Art, Archimedes 37,

DOI 10.1007/978-3-319-05065-2_2, © Springer International Publishing Switzerland 2014

mailto:Sylvie.Neven@ulg.ac.be

writings describe processes for the manufacture, preparation and application of

various types of materials and substances. The majority are anonymous compi-

lations of texts, which may originate from older or undetermined authorities.

Hundreds of such collections of recipes dealing both with alchemical and

art-technological procedures were produced and disseminated in Northern Europe

from the fourteenth century on, especially in German-speaking countries.

Drawing on a delimited corpus of about 40 representative German manuscripts dated from the fourteenth to the sixteenth century, this paper investigates the connections and similarities between these two fields and examines the various ways alchemical and artistic instructions were embedded within recipe books.1 It

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
literature quoted there.
16 Newman, Promethean Ambitions, 115–63.
17 Principe & DeWitt, Transmutations, 11–27.
18 For experts and entrepreneurial alchemy, see Nummedal, Alchemy and Authority, 40–5.

It is in this context of alchemical expertise that we should consider the artisans

who adopted the language of alchemy in the early modern period to attract the

attention of patrons. One example can be seen in the description of Black Berthold,

the legendary discoverer of gunpowder in Europe, in the late fifteenth-century

manuscript Feuerwerckbuch. Depicted in the company of a furnace and alembics,

Berthold’s creation of ordnance is identified with alchemy. Black Berthold, the

Feuerwerckbuch reads,

worked with the great alchemy like those masters who are engaged with precious and

valuable things, with silver and gold, and with similar metals. These masters can separate

silver and gold from other precious jewelry, and from other valuable colors which they can

produce. Now this master Berthold wanted to induce a golden coloration. For this

he used

salpetre, sulphur, lead and oil. Then he put these ingredients in a container made of copper,

which he sealed completely, exactly as it should be done, but when he put it on fire and the

container became hot, it burst into many pieces.19

The Feuerwerckbuch witnesses material production based on practical chemistry in

terms borrowed from alchemical transmutation, or more precisely, the making of

gold. The analogy was not always with chrysopoeia as such. Vasari’s already

mentioned fashioning of Van Eyck as an alchemist shows that the image of alchemy

as a field of expert inquiry on materials and material transformation was considered

sufficiently positive to identify with. However, we should not forget that Vasari's

identification of Van Eyck as an alchemist served the purpose of downplaying

Netherlandish art as techne, only a first step in a narrative of art historical progress

culminating in Vasari's beloved Florentine art.

The epistemic value of techne was shifting at the time of Vasari. Artisans came

to be considered experts of nature likening the artisanal processes of material

transformation undertaken in their workshops to those of nature.20 As Andrea

Bernardoni shows in this volume, Biringuccio's Pirotechnia is part of this larger

process of shifting epistemic value of artisanal processes. Rejecting

transmutational

alchemy as “false” and the alchemists who practiced it as fraudulent, Biringuccio

carved out space for true alchemy as one of the arts of fire. Artisanal workshops,

such as Andrea Verrocchio’s (c.1435–1488) in which Leonardo da Vinci (1452–

1519) apprenticed or Leonardo’s own workplace, shared a material culture with

alchemical laboratories. As we have already pointed out, artisans used similar

equipment and performed “chemical” operations. However, Bernardoni argues,

Biringuccio made the claim that these artisans were the true experts on matter,

materials and material transformation and that artisanal “chemical” operations were

the key to natural knowledge.

One of the readers of Biringuccio’s Pirotechnia was Johannes Mathesius (1504–1565), a Lutheran preacher in St. Joachimsthal, the center of an important mining district. In her contribution to this volume, Henrike Haug analyzes Mathesius’s

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
William Newman, Tara Nummedal, Doris Oltrogge, Valentina Pugliano, Jennifer

Rampling, Marlise Rijks, Maurice Yves-Christian Sass, Lorenz Seelig, Pamela

Smith, Anke Timmermann, Barbara Tramelli, Berit Wagner, Arie Wallert, Ittai

Weinryb, Steve Wharton, and Alan Williams.

This book would not have been possible – and even less possible in such a short

time – without the efficient and magnificent editorial assistance of Gina Partridge

Grzimek, Lea Poeschl and Celine Camps. My final thanks go to Jed Buchwald for

his trust in the project and for considering the book for his series Archimedes, and to

Lucy Fleet for her assistance, advice and patience throughout the several stages of

this edited book project.

Berlin, Germany Sven Dupré

Bibliography

Beretta, Marco. 2009. The Alchemy of Glass. Counterfeit, Imitation and Transmutation in
Ancient Glass-Making. Sagamore Beach: Science History Publications.

Davis, Lucy. 2009. Renaissance Inventions: Van Eyck’s Workshop as a Site of Discovery and

Transformation in Jan van der Straet's Nova Reperta. In Envisioning the Artist in the
Early Modern Netherlands. Netherlands Yearbook for History of Art 59, eds. H. Perry Chap-

man and Joanna Woodall, 223–248. Zwolle: Waanders Publishers.

Dupré, Sven. Forthcoming. The Value of Glass and the Translation of Artisanal Knowledge in

Early Modern Antwerp. In Trading Values, Netherlands Yearbook for Art History, eds. Bart
Ramakers, Christine Göttler, and Joanna Woodall.

Evans, R.J.W. 1984. Rudolf II and his World: A Study in Intellectual History, 1576–1612.
Oxford: Oxford University Press.

Ganzenmüller, W. 1956. Das chemische Laboratorium der Universität Marburg im Jahre 1615. In

Beitr€age zur Geschichte der Technologie und der Alchemie, ed. W. Ganzemüller,

314–322.

Weinheim: Verlag Chemie.

Göttler, Christine. 2013. The Alchemist, the Painter, and the “Indian Bird”: Joining Arts and

Cultures in Seventeenth-Century Antwerp. Adriaen van Utrecht’s Allegory of Fire in the

Royal Museums of Fine Arts in Brussels. In Synergies in Visual Culture. Bildkulturen im
Dialog. Festschrift f€ur Gerhard Wolf, eds. Manuela De Giorgi, Annette Hoffmann, and

Nicola Suthor, 499–512. Munich: Wilhelm Fink.

Hannaway, Owen. 1986. Laboratory Design and the Aim of Science: Andreas Libavius versus

Tycho Brahe. Isis 77: 585–610.

Hassenstein, Wilhelm. 1943. Das Feuerwerckbuch von 1420: 600 Jahre deutsche Pulverwaffen
und Buchsenmeister. Munich: Deutschen Technik.

Howard, Rio. 1983. La Bibliothèque et le laboratoire de Guy de La Brosse au Jardin des Plantes à Paris. Genève: Droz.

Joly, Bernard. 1992. Qu’est-ce qu’un laboratoire alchimique?. Cahiers d’histoire et de philosophie des sciences 40: 86–102.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
19 Hassenstein, Das Feuerwerckbuch, 45–6. For translation and discussion, see Werrett,

Fireworks, 28.
20 Smith, Body of the Artisan. esp. 95–127.

sermons to reveal conceptions of the origin and formation of ores for which

Mathesius drew equally on natural philosophy and alchemy and on the artisanal

knowledge of the local miners and goldsmiths. Following one of the goldsmiths’

most prized objects, a so-called Handstein, into the Kunstkammer, Haug shows that
this knowledge also reached elite collectors. They valued Handsteine for the

metallogenetic knowledge they embodied, thereby endorsing the shifting epistemic

status of the arts.

Epistemic Changes Between Artisans and Alchemists

Biringuccio and Mathesius’s St. Joachimsthal are examples of persons and places

of exchange between scholarly cultures (in which learning is based on reading and

writing) and artisanal cultures (in which learning is based on doing).21 The labo-

ratories created in Medici Florence, discussed in the chapters by Kieffer and

Beretta, were also early examples of such places of exchange, followed by many

other European courts. These workplaces facilitated the exchange of materials

between alchemy and the arts, and the attraction of the courts also made artisans

adopt alchemical language to elevate the status of their craft. However, above all,

these court laboratories made possible, well beyond a shared material culture, the

exchange of people and knowledge between the arts and alchemy. Although there is

a long tradition of experimentation in alchemy and the boundaries between alchemy

and art technologies were fluid from the very beginnings of alchemy in Antiquity, in

the sixteenth and seventeenth centuries artisans became more deeply involved in

alchemical pursuits, and some crafts relied on chemical expertise offered by scholars trained as alchemists.22 Above all, texts and books, products and symbols of scholarly culture, played an increasingly important role in laboratories and workshops. In these workplaces, a sort of hybrid figure was at work, with one foot in artisanal culture and another in scholarly culture and impossible to catego-rize in mutually exclusive categories of the scholar and the craftsman.23 Certain types of crafts—glassmaking, gold- and silversmithing, and porcelain production—seem to have been particularly prone to exchanges between artisanal and scholarly alchemical cultures. By the seventeenth century, the expertise of some glassmakers,

silver- and goldsmiths and producers of porcelain was just as based in the worlds of

alchemical and bookish learning as it was grounded in hands-on work in the

laboratory.24 Lawrence Principe and Morgan Wesley discuss two examples of

such arts: silversmithing and porcelain production, respectively.

21 For ‘trading zones’ between artisanal and scholarly cultures, see Long, Artisan/Practitioners.
22 Klein, “Chemical Experts.”
23 On hybrid experts, see Klein, “Artisanal-Scientific Experts,” and “Depersonalizing the Arcanum.”
24 For glassmaking (not discussed here), see Beretta, Alchemy of Glass; and Dupré, “Value of

Glass.”

Silver- and goldsmiths seem obvious candidates when we think of careers that

cross artisan and alchemist. Silversmiths, goldsmiths and alchemists

worked on the

same materials (silver and gold), and silver- and goldsmiths’ expertise was called

for to assay the purported gold and silver transmuted by alchemists. Silver- and

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
argues that textual form and lexical proximities within recipes from these different

disciplines may lead to association, contamination and confusion within this textual

genre. It finally suggests some clues to help locate, distinguish and demarcate

alchemical content within the literature of recipes.

Art and Alchemy Within Recipe Books

At first sight, any overlap between alchemy and art-technology within recipe books

can be broadly explained by the mutual use of various materials and substances

such as “common stones, gems, and types of marble, gold and other metals, sulfurs,

salts, and inks, azures, minium, and other colors, oils and burning pitches, and

countless other things.”2 More precisely, the field of art-technology encompasses a

large range of craft practices involved in the production of pieces of art (including

those which incorporate such substances). This ‘hand’ knowledge, is related to the

mechanical arts and is divorced from the philosophical or speculative dimension.

Yet, alchemy could be described as the practical, philosophical and medical search

for the perfecting of base material substances and also for the extending of life.3

The theoretical and practical aspects of alchemy involve both the study of all inanimate or animate things made from the elements and the observation and imitation of natural processes within the laboratory.4 In this context, alchemy could be seen as a mechanical art, in the sense that it works on matter but is also a liberal art, as it attempts to explain matter in its composition and its transformation.5 In the practical sense, one of the main goals of alchemy is the transmutation

1 The main data and characteristics of these manuscripts are given in Appendix.
2 Brewer, Fr. Rogeri Bacon, 39–40.
3 Perception and definition of alchemy is not chronologically constant and has been the subject of

several (re)interpretations since the eighteenth century, see Principe & Newman, “Historiography of Alchemy.”

4Halleux, “Alchimie,” 336–7; Newman, “Technology and Alchemical Debate,” 432–3; Pereira, “Use of Vernacular Languages,” 336; and Kahn, Alchimie et paracelsisme, 7–8.

5 Halleux, Savoir de la main, 134.

of base or impure matter (often metal or stone) into a noble or perfect one.6 To do so, alchemists used to perform chemical processes and manipulations which resembled those practiced by contemporary artists and artisans.

Thus, in both fields particular importance is placed on craft practices. Both alchemical and artistic recipe books describe various processes for purifying and

transforming materials, either for improving their properties or in order to use them

for specific purposes. In this context, from a technological point of view, the term

‘transmutation’ could refer to the colouring of glass, the melting and tinting of

metals, the dyeing of stones or gems, or the manufacture of synthetic pigments. A

huge number of recipes are dedicated to procedures for obtaining gold or silver or

gold-silver like substances from base metals (copper, tin, lead, iron, mercury).

The tradition of recipe books has roots deep in Antiquity. Treatises dealing both

with art-technological and alchemical procedures notably survived in the Alexan-

drian Greek papyri preserved in Leyden and Stockholm.7 These date from the

third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:

Fanny Kieffer

Material and Temporal Powers at the Casino di San Marco

(1574–1621) 129

Marco Beretta

Goldsmiths and Chymists: The Activity of Artisans Within

Alchemical Circles 157

Lawrence M. Principe

Igne refutata: Thermal Analysis in the Laboratory Practices

of John Dwight and Ehrenfried Walther von Tschirnhaus 181

Morgan Wesley

ThiS is a FM Blank Page

Introduction

The Antwerp painter Adriaen van Utrecht (1599–1652) arranged Chinese porce-

lain, a goblet made of rock crystal and cristallo glasses à la façon de Venise, next to
a magnificent display of gold- and silversmiths’ works on a table in the foreground

of his 1636 “Allegory of Fire” (Fig. 1).1 Many of these objects have been identified

as originating in Antwerp, and it is likely that Van Utrecht’s painting celebrates

the manufacture and trade of luxury goods for which Antwerp gained fame in the

early seventeenth century. Van Utrecht’s objects have another point in common:

one way or the other they are produced through the agency of fire. Van Utrecht’s

painting celebrates Antwerp’s economic power and the productive ingenuity of its

craftsmen and artists as much as it praises the transformative force of fire turning

relatively cheap and humble materials into highly valued objects of art. These

objects were all products of the arts of fire, which according to Vannoccio

Biringuccio (1480–c.1539) included alchemy, and excluded “false alchemy”

concerned with pretentious transmutation.2

In the background of Van Utrecht’s painting, a window opens on to a space

in

which a man stands working at a stove with an open fire. Stirring a cauldron, the

man is shown in the material company of bellows, an anvil, a melting and a

distilling furnace, and other equipment related to the worlds of assaying and

metallurgy. Van Utrecht’s background refers to the spaces in which the objects in

the foreground were produced. Were these spaces laboratories or artisanal work-

shops? Were they home to gold- and silversmiths, glassmakers or producers of

porcelain? The difference between workshops and laboratories during this period is

ambiguous, not only in Van Utrecht’s depiction, and the line between the two

spaces is as difficult to draw as that between the arts of fire and alchemy. This book

is concerned with the interconnections and differentiations between foreground and

1My discussion of Van Utrecht’s painting in this and the next paragraph is based on Göttler,

“The Alchemist, the Painter.”
2 For Biringuccio’s attitude towards alchemy, see Newman, Promethean Ambitions, 128–32.

background in Van Utrecht’s painting, between artisanal workshops and alchemical

laboratories, between the material arts and alchemy.

Van Utrecht is one of the heirs of Van Eyck (c.1390–1441). Following Giorgio

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Omnifaria Doctrina) and some specific essays (see Bidez, Épitre sur la Chrysopée, 51–70; and
Duffy, Michaelis Pselli, texts 19–31). On the relations between Psellos’s alchemical interests and

his investigation of the physical world, see Katsiampoura, “Transmutation of Matter,” 665–7.
4 “τὸ μὲν γὰρ ὕδωρ πηγνύμενον λιθοῦται ει$ ς κρύσταλλον [and] τοῦτο ει$ ς ἀτμίδα λυόμενον ἀὴρ καθίσταται.”

5 In § 4 Psellos explains the petrifaction of an oak struck by a lightning strike: a quick and sharp lightning does not only make the oak black, but consumes all its humidity and transforms the wood into stone.

6 See infra, § 3.

7 The entry continues by telling how the emperor Diocletian (284–305) made to burn all the Egyptian books on alchemy (in the Greek text: “περὶ χημείας χρυσοῦ και ἀργύρου”); this information probably depends on the Chronicle (Ἱστορία χρονική) of John of Antioch (active under Heraclius’s reign, 610–641): see fragment 280 in Roberto, Ioannis Antiocheni Fragmenta.

Empire, the same definition had been already given by the Persian scholar al-Nadīm

(ninth century AD), who devoted the tenth chapter of his Kitāb al-Fihrist (Book of
Catalogue) to the s
.

ināʻat al-kīmīyā’ (art of alchemy): at the very beginning of the

chapter this art is defined again as “the making of gold and silver from other

metals.”8 More than one century later, the Syriac scholar Bar Bahlul (eleventh

century AD) dealt with the same subject in two entries of his lexicon:

1. Duval, Lexicon Syriacum, vol. I, 901: Kimiya, the dark stone. It is the work of the art of gold and silver. And there is

someone who explains this word from the name kima (the constellation of the

Pleiades), that is the height stars, because (this art) is accomplished by means of

the height mixtures.9

2. Duval, Lexicon Syriacum, vol. I, 904: Kemela’a, that is the alchemy (kimiya’a) of the seven bodies and of the various natures; [it includes] arts such as goldsmith’s art and copper, iron, and glass working.10

Only the last definition includes a wide set of techniques applied to different substances (among which glass is mentioned), while most of the above-quoted texts identifies alchemy with the making of gold and silver. The two precious metals were probably the most attractive outcomes that anyone engaged in alchemical

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Matteo Martelli shows in this volume, relatively early on a historical process of

selection, appropriation and differentiation resulted in a more limited definition of

Fig. 2 Pieter Brueghel the Elder, The Alchemist’s Family: Al-gemist, 1558 (Courtesy of the

Kupferstichkabinett, Staatliche Museen zu Berlin)

14 For the scope of early modern alchemy and the notion of ‘chymistry’, see Principe, Secrets of
Alchemy; and Newman & Principe, “Alchemy vs. Chemistry.”

alchemy primarily focused on the making of gold and silver. Already around

300 AD, a distinction was introduced between a limited definition of alchemy as

metallic transmutation and a more encompassing definition including productive

knowledge and various artisanal technologies.

Transmitted to Europe, and translated into Latin and the vernaculars, several of

the recipes in the Papyri are still found in collections of recipes in the fifteenth and

sixteenth centuries. More importantly, as Martelli remarks in his essay, the scope of

a recipe collection such as the Mappae clavicula, compiled between the ninth and

the twelfth centuries, is as encompassing as that of the Papyri despite the earlier

attempts to limit alchemy to metallic transmutation only. It should not come as a

surprise then that Sylvie Neven finds it difficult to demarcate between alchemical

and art technological recipes in late medieval collections of recipes. Her contri-

bution to this volume shows that alchemical and art technological recipes shared a

concern with the same materials and artisanal processes. The focus on chromatic

transformation already found in the Papyri continues in fifteenth-century recipes.

Although related to laboratory practices, Neven emphasizes that these recipe

collections are the products of scribal compilation and copying processes. Words

and works are equally important elements of alchemical practices.15 Next to

laboratories, medieval religious institutions were also important sites of alchemical

and art technological practice. However, as Neven points out, this does not exclude

that some scribes, such as the Benedictine monk Wolfgang Seidel (1491–1562),

tried out recipes.

The scope of alchemy was from its very beginnings contested, and remained so

throughout its long history. Especially its boundaries with art technology were fluid.

Throughout the Middle Ages and the Renaissance, the distinction between

alchemy

as strictly goldmaking and a more encompassing definition overlapping with art

technology was crucial to the polemics of artists and alchemists and the rivalry

between alchemy and the arts.16 Given the contested nature of the field of inquiry,

it follows that the identity of the alchemist was equally contested and complex. In

the early modern period the alchemist was often portrayed as a fraud. The portrayals

of laboratory scenes building on Brueghel became a genre of its own in the

Netherlands.17 In these scenes, the alchemist is a goldmaker searching in vain for

the Philosophers’ Stone and riches. As Tara Nummedal has convincingly shown,

the portrayal of the alchemist as a fraud also created the opportunity for other

alchemists to fashion themselves in the role of experts, offering their services to

courts and other patrons to help them unmask fraudulent alchemists.18 Alchemical

expertise was based on a broader knowledge of matter and materials.

15 For reading and writing as alchemical practices, see Nummedal, “Words and Works,” and the

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Art Technologies and Knowledge of Material Transformation

Laboratories were not only the workplaces of transmutational alchemists. TheUffizi
and the Casino had little in common with the workplaces of alleged goldmakers

destined to fail and bring their families to financial ruin, as famously depicted by

Pieter Brueghel the Elder (c.1525–1569) (Fig. 2). Just as the spaces known as

laboratories encompassed the workplaces of artisans laboring at furnaces and

using materials and chemical procedures to make both everyday items and objects

of the visual and decorative arts, the activity known as alchemy encompassed more

than attempts to make gold.14 Transmutational alchemy was about the trans-

mutation of all base metals into more noble ones, but chrysopoeia was only one

aspect of alchemy. Alchemy also touched on medicine and chemical manufacture.

It was about the chemical production of things—medicines, porcelain, dyes, and

other products as well as the precious metals—and about the knowledge of how to

produce them. In this sense, “art technologies”—materials and techniques to make

art and knowledge of these materials and techniques—overlapped with alchemy.

Alchemy has deep roots in writings on material transformation from Antiquity.

The productive knowledge associated with material transformation was written

down in recipe books. The Leiden and Stockholm Papyri date to the third century

AD. They contain recipes for the making of gold and silver, for the imitation of

precious stones, and for textile dyes. Whether a recipe is about coloring silver to

make it look like gold or the making of an imitation ruby or another artificial

precious stone, all of the recipes in the Papyri are, indeed, about material trans-

formation and color change as related to the manufacture of luxury goods. As

Matteo Martelli shows in this volume, relatively early on a historical process of

selection, appropriation and differentiation resulted in a more

limited definition of

Fig. 2 Pieter Brueghel the Elder, The Alchemist’s Family: Al-gemist, 1558 (Courtesy of the

Kupferstichkabinett, Staatliche Museen zu Berlin)

14 For the scope of early modern alchemy and the notion of ‘chymistry’, see Principe, Secrets of
Alchemy; and Newman & Principe, “Alchemy vs. Chemistry.”

alchemy primarily focused on the making of gold and silver. Already around

300 AD, a distinction was introduced between a limited definition of alchemy as

metallic transmutation and a more encompassing definition including productive

knowledge and various artisanal technologies.

Transmitted to Europe, and translated into Latin and the vernaculars, several

of

the recipes in the Papyri are still found in collections of recipes in the fifteenth and

sixteenth centuries. More importantly, as Martelli remarks in his essay, the scope of

a recipe collection such as the Mappae clavicula, compiled between the ninth and

the twelfth centuries, is as encompassing as that of the Papyri despite the earlier

attempts to limit alchemy to metallic transmutation only. It should not come as a

surprise then that Sylvie Neven finds it difficult to demarcate between alchemical

and art technological recipes in late medieval collections of recipes. Her contri-

bution to this volume shows that alchemical and art technological recipes shared a

concern with the same materials and artisanal processes. The focus on chromatic

transformation already found in the Papyri continues in fifteenth-century recipes.

Although related to laboratory practices, Neven emphasizes that these recipe

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
The Alchemical Art of Dyeing: The Fourfold Division of Alchemy. . . 3

Empire, the same definition had been already given by the Persian scholar al-Nadīm

(ninth century AD), who devoted the tenth chapter of his Kitāb al-Fihrist (Book of
Catalogue) to the s
.

inā‘at al-kīmīyā’ (art of alchemy): at the very beginning of the

chapter this art is defined again as “the making of gold and silver from other

metals.”8 More than one century later, the Syriac scholar Bar Bahlul (eleventh

century AD) dealt with the same subject in two entries of his lexicon:

1. Duval, Lexicon Syriacum, vol. I, 901: Kimiya, the dark stone. It is the work of the art of gold and silver. And there is

someone who explains this word from the name kima (the constellation of the

Pleiades), that is the height stars, because (this art) is accomplished by means of

the height mixtures.9

2. Duval, Lexicon Syriacum, vol. I, 904: Kemela’a, that is the alchemy (kimiya’a) of the seven bodies and of the

various natures; [it includes] arts such as goldsmith’s art and copper, iron, and

glass working.10

Only the last definition includes a wide set of techniques applied to different

substances (among which glass is mentioned), while most of the above-quoted

texts identifies alchemy with the making of gold and silver. The two precious metals

were probably the most attractive outcomes that anyone engaged in alchemical

practices tried to achieve out of this art, as is possible to infer from a few accounts

on alchemists preserved by Byzantine and Syriac chronographers. In the mid-sixth

century, for instance, John Malalas (491–578) reported the story of John Istmeos,

χειμευτὴς ὑπάρχων καὶ φοβερὸς ἐπιθε﹖της (alchemist and tremendous impostor),

who moved from Antioch to Constantinople, where he was condemned by the

emperor Anastasios (491–518) because of the adulterated gold objects he tried to

8 I have quoted the translation proposed by Dodge (The Fihrist, 841), who checked several

manuscripts (see pp. xxiii–xxxiv) not taken into account by Flügel, who edited the Arabic text.

In Flügel's edition (Kitāb al-Fihrist, vol. II, 351) the passage reads:
, translated by Fück, "Arabic Literature of Alchemy," 88, as follows:

"The adepts of the Art of Alchemy, which is the art of making gold and silver without (recourse to)

mining."
9

The expression literally

means 'stone of sadness.' Berthelot & Duval, Chimie, 133 translated it as 'pierre philosophale.'

Instead of ('sadness, grieving'), I read ('black, dark'); on the tendency of

identifying ‘alchemy’ with the substance used for transforming vile metals into gold and silver,

see Gildemeister, “Alchymie.”

10

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
philosophy and alchemy and who perhaps conducted their own experiments, as

suggested by formulae which follow some of the recipes, such as probatum vom Bischoff von Freising (Cgm 4117, fol. 2v). Schobinger is notably at the root of a

large compilation of alchemical texts.32 He is also renewed for having personally

known Paracelsus, who referred to Schobinger’s writings.33 The value of such an

authority may appear visually in the recipe book. In the Cgm 4117, Seidel dedicates

a whole page to recording Schobinger's name.34 Moreover, the simple invocation

28 See, for example, Barthélemy, "Alchimie et médecine," 110–3.
29 Paulus, "Wolfgang Seidel"; and Pöhlein, Wolfgang Seidel.
30 "So vill vom geschenckh hab ich auss der liberej des closters zw sant vlrich zw Augspurg lassen

abschreiben durch ain knaben des namen ist Walthasar Gech von Fiessen im 1550 Jahr." (Munich,

Bayerische Staatsbibliothek, Cgm 4118, fol. 128r).
31 Schobinger, Schowinger von St. Gallen.
32Allgemeine Deutsche Biographie, 209; and Hertenstein, Joachim von Watt, 91–2.
33Meier, Paracelsus, 33–46.
34 "Von bartholome Schobinger burger zu sanndt Gallen in Schweitz. Hab ich dise nachuolgende

kunstel. etc./Empfangen den Sibenvnndzwaintzigisten tag. des Monats Februarii/Anno etc. 40.”

(Munich, Bayerische Staatsbibliothek, Cgm 4117, fol. 62r).

of the name of the Bishop of Freising would have served to confirm the efficacy of

some of the technical instructions. Thus, the same way the scribes used to relate old

treatises or data with the name of previous and quoted authorities, such as (pseudo)

Albertus Magnus or Arnaldus de Villa Nova, they also mention those of their

contemporaries to lend authority to validate the practicability or the reproducibility

of the instructions they consign.35

In some cases, the information recorded in recipe books is documented as having

been provided by an artist or practitioner. Augsbourg Staats- und Stadtbibliothek 2

Cod. 207 was produced in St. Ulrich and St. Afra's Cloister. It contains miscella-

neous alchemical treatises and collections of recipes contributed by several scribes,

including the monk Bild Vitus (1481–1529) and Johannes Gossolt (1421–1506),

identified as vicarius augustensis.36 In this work, Gossolt combined alchemical

treatises attributed to (pseudo) Albertus Magnus with Latin and German alchemical

recipes. For the latter he sometimes specifies his local sources. For example, at folio

171v, he mentions the "Magistri Jodoci Aurifabri de Haidelberga." Other citations

of goldsmiths' names are found in our corpus of texts. In the St. Gallen Cod.

Vadiana 395, several alchemical instructions are associated with the name of

“Nicolaus Aurifaber.” In many respects, metalworkers seem to have shared interest

and knowledge in alchemical practices and materials.37

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
different recipes.

As stated above, whether alchemical or art-technological, the recipes contained

in these manuscripts are presented in the form of a formula which, in most cases,

enumerates the ingredients and the actions necessary to produce a particular

preparation. In addition artistic recipes sometimes indicate the recommended

geographical provenance or grade of quality of the ingredients. Suggestions for possible substitutions might also appear. This sort of information is rare in alchemical recipes.

The length of a recipe depends not only on the number of ingredients involved but also on its complexity, the number of steps necessary to obtain the final product.

A recipe can be anything from one sentence to several pages within a manuscript.

Alternatively, a recipe may appear merely as a brief list of ingredients, without any other additional information. In fact, two categories of recipe can be distinguished: the Vollrezepte (detailed recipes) and the Kurzrezepte (abbreviated recipes).55 In

the first, the quantities and the various steps are indicated. In the second, only the

55Halleux, “Alchimie,” 343.

ingredients are cited: the procedure is sketched out or omitted altogether and the

rest is left to the ingenuity of the user. This second category is more common in the

case of artistic recipes; a great many of the recipes dedicated to the manufacture of

ink are written in the form of a very short list of ingredients. It is less common for

the alchemical recipes to be presented this way.

The title of a recipe may also give an indication of the final product to be obtained

and, in some cases, specify the use of the product. Again, this is particularly true for

artistic instructions and is less observable for alchemical ones.

For both types of instructions (alchemical or artistic), some steps could be

omitted or were left to the interpretation of the reader. Specified quantities may

be missing in both fields. When quantities are given, artistic recipes are far more

likely to use local measurements, whereas in alchemical instructions, the quanti-

ties—if mentioned at all—are more often expressed in terms of ratio or proportions.

In some cases, these proportions are not ‘practically’ correct. A very well-known

example is the proportion of mercury and sulphur proposed by mediaeval recipes

for the production of vermilion which is invariably incorrect.56 Very rarely are the correct chemical proportions cited.

In parallel, alchemical writings may involve the use of symbols or metaphors to designate substances and practices. In consequence, the way an alchemical recipe was received would depend on the degree of experience of the reader-practitioner reading it. On the one hand, the (sometimes) metaphorical or codified language as well as the approximations stressed the arcane nature of these recipes and contributed to their secrecy. On the other hand, the omitted information may have been complemented by data only known to some readers and not recorded by the copyist who conserves only the essential part of the recipe. If so then this kind

of

recipe was only meant to be accessible and useable by those practitioners who could

easily fill in the lacuna that punctuated the text of the recipe.

As previously observed, citations of authority were frequently used by the scribes

of the manuscripts. However, the tendency for an older authority to be cited in the

recipe books is particularly characteristic of the alchemical writings and less typical

of the art-technological recipes. As stated above, such citations primarily served to

legitimate the technical and chemical procedures. In addition, most alchemical

Visible: 0% - 95%

ASSISTANT

Here's a list of mentioned across *Laboratories of Art: Alchemy and Art Technology from Antiquity to the 18th Century* with notes on how they relate to and . I've grouped them into categories (metals, stones, pigments, dyes, chemicals, organics) to reflect how alchemists and artisans worked with them:

---

1. – Goal of chrysopoeia; gilding, chrysography, luxury art【8:3†source】.
2. – Used in alchemy, coinage, and silvering processes【8:3†source】.
3. – Base for alchemical transmutations; alloys (bronze, brass)【8:5†source】.
4. – Common "base" metal in recipes, often alloyed【8:5†source】.
5. – Associated with Saturn; used in pigments and metallurgy【8:5†source】.
6. – Base metal, forging, dyeing stones,

tools【8:5†source】.

7. – Central to alchemy (mercurial principle), used in gilding【8:11†source】.

8. – Used for alloys, glass coloring, medicine【8:6†source】.

9. – Early modern metallurgy, imitation of precious metals.

10. – Artistic and alchemical alloy of copper & tin.

---

11. – Fused to make cristallo glass【8:3†source】.

12. – Carving, dyeing in recipes【8:4†source】.

13. – Used for imitation gems【8:4†source】.

14. – Recipes for green stones【8:3†source】.

15. – Recipes for imitation gems【8:15†source】.

16. – Glass recipes

mimic sapphire hues.

17. – Mentioned in gemstone imitations.

18. – Ground for pigments, imitated with glass.

19. – Target of imitation recipes (artificial stones).

20. – Collected by goldsmiths for knowledge of metals 【8:10†source】.

---

21. – Mineral pigment, linked to alchemy 【8:4†source】.

22. – Luxury pigment, costly and alchemically significant.

23. – Artificial pigment, linked to lead 【8:4†source】.

24. – Artificial pigment from mercury + sulfur 【8:18†source】.

25. – Ink and paint, made from fire.

26. – Earth pigments, transformed by heat.
27. – Yellow pigment, alchemically dangerous.
28. – Red/orange pigment, also alchemical reagent.
29. – Green pigment, copper ore derivative.
30. – Ancient dye, referenced in papyri【8:5†source】.

---

31. – Base substance in recipes【8:4†source】.
32. – Used in gunpowder, fireworks, and alchemy 【8:8†source】.
33. – Alchemical principle; pigments, gunpowder【8:8†source】.
34. – Used in dyeing, acids, and metallurgy.
35. – Mordant for dyes and pigments.
36. –

Used in metalworking.

37. – Flux in glass and metallurgy.

38. – For whitening metals, pigments, alchemical “poison.”

39. – In recipes for preservation and colors.

40. – Mortars, fresco, and glass production.

---

41. – Paint binders, varnishes【8:1†source】.

42. – Protection of art, linked to painter’s “alchemy.”

43. – Burning pitches in recipes【8:4†source】.

44. – Used in casting, encaustic painting.

45. – Binding agents in inks and paints.

46. – Confections, alchemical purifications, sugar sculpture【8:3†source】.

47. – Distillation in both

alchemy and artisanal recipes.

48. – For dyes, medicines, and alchemical tinctures.

49. – Dyed in recipes, parchment, artisanal uses【8:5†source】.

50. – Recipes for writing, both artisanal & alchemical 【8:4†source】.

📌 The book shows how

. Alchemists sought transmutation and perfection, while artisans used the same substances in practical crafts—glassmaking, gilding, pigment manufacture, dyeing, porcelain, fireworks, and metallurgy. Recipe books often mixed instructions for both domains, blurring boundaries 【8:4†source】【8:15†source】.

Do you want me to

with columns for *Material*, *Alchemy Involvement*, and *Artisanal Culture* (like

pigments → transmutation vs. painting), so you get a clear comparative view?

USER

continue with any materials we've missed

ASSISTANT

{"queries": ["list of substances, ingredients, or materials used in alchemy or artisanal practices", "materials mentioned in dyeing, glassmaking, porcelain, or metallurgical recipes", "chemicals and substances in alchemical laboratories", "materials in the Uffizi and Casino di San Marco workshops", "ingredients for pigments, dyes, or artificial gems in alchemy"]}

TOOL

[Web browsing]

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
equipment from the Casino to the Uffizi. During Francesco I’s reign the Casino housed a very important artistic glass workshop where the prince worked on his

own experiments (for example smelting rock crystal).29 This laboratory was lead by

Sisti, who also lead a glassmaking laboratory in Pisa, where he often had to go. In

his early career he used the technique a lume di lucerna, but once he transfered

to

the Uffizi, this activity became secondary.30

At the Uffizi, glass production may well have been located in the smithy

workshop. It was probably limited to the glass technique a lume di lucerna. The oven glass manufacturing method came from existing production centres such as

the ones in Pratolino, or Pisa, which became the most important in Tuscany. On the

other hand, during Ferdinando I’s reign, documents already show activities by

someone called Niccolò di Vincenzo Landi di Lucca, after Sisi, the principal

glassmaker in service to the Medici’s till 1620. The sources show that Landi, a

specialist in a lume di lucerne, went on duty in the fonderia of the Uffizi in 1591,

where he made little animals for the decoration of glass manufactured in

Pisa.31 In

1601, Antonio Neri also mentioned him as the leader of the new glasswork in the

Casino di San Marco.32 Later, in 1618, when Grand Duke Cosimo II (1590–1621)

set up a large workshop with many ovens in the Boboli gardens near the Pitti palace,

28 ASF, GM 403, ins. 2, fol. 120.
29 On the story of the Medicean glassmaking, see Heikamp, “Mediceische Glaskunst.”
30 The technique a lume di lucerna allows to make or decorate little objects heating locally the

glass elements thanks to a lantern flame or a candle.
31 ASF, GM 112, passim; GM 217, fol. 23; and GM 195, ins. 1, fol. 102.
32 On the priest Antonio Neri and his treatise on the art of glassmaking, see Abbri’s introduction to

Neri, L’arte vetraria (2001), 5–23.

he entrusted its management to him, which shows that he was certainly able to

supervise large-scale production.

The fonderia produced mainly small ornamental glass items (Fig. 7). The items

in use for distillery and for the other workshops came either from Pisa, or from the

Casino di San Marco.33 Mainly made of bronze or metal, the models of these little

items were created by the court goldsmiths or by the smelters: there were little

masks, buttons to be filled with perfumes, ornamental glasses.34 The glasses, stored

in a hall of the Galleria “la stanza dei cristalli,” were distributed according to the

grand duke’s wishes and other workshops requirements (most of them were given

as presents or used to decorate other items coming from other workshops).35 Sisti,

in charge of the fonderia, was responsible for the seamless transitions between the

different stages of the process, which was not always an easy task in the case of a

delicate material such as crystal.36

The Forge

In the Uffizi, the forge and smelting works were far less visible and have less

prestige. Only small items are worked on in the forge: little masks for the orna-

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
application of its techniques to reverse the process of making porcelain is a logical

extension of the economic demand for china and the intellectual drive toward

maker’s knowledge as expressed by Bacon.

Certainly, the history of ceramics after the seventeenth century is inextricably

linked to chemistry’s success at solving increasingly sophisticated questions of

luxury pottery production. We need look no further than William Cookworthy’s

(1705–1780) patent for the manufacture of porcelain in 1768 or the thousands of experiments conducted by Josiah Wedgwood (1730–1795) in the production of his Jasperwares. Unlike the other arts of the fire; metalworking, glassmaking, and dyeing, the late inclusion of ceramics into the hierarchy of decorative arts allows contemporary scholars opportunities to investigate the separations between the craft artisan and the chymist during the early modern period, of which this discus-sion of thermal analysis is only one of many.

Table 1 A comparison of modern elemental analyses of ‘Dwight’ experimental ware, Blanc-de-Chine, ‘Böttger’ ware, and Kangxi porcelain

Material	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	Other	Total

Dwight 80.3 12.4 0.74 0.39 0.2 5.35 0.47 Trace 99.1

Blanc-de-Chine (ft46) 76.7 16.8 0.35 0.15 0.08 5.9 0.2 Trace 100

Kangxi Porcelain (ft47) seven-

teenth century

64.7 28.35 0.95 0.5 0.1 2.8 2.4 0.19 99.99

Böttger Porcelain 1708 61.0 33.0 0.00 4.8 0.00 0.1 0.2 0.9 100

Meissen Porcelain (ft48)

mid-eighteenth century

59.0 35.00 0.00 0.3 0.00 4.0 0.00 0.9 100

Bibliography

Bacon, Francis. 1858. New Organon, trans. James Spedding et al. London: Longmans and Co.

Birch, Thomas. 1756–1757. History of the Royal Society of London. London: A. Millar.

Biringuccio, Vannoccio. [1540] 1942. The Pirotechnia of Vannoccio Biriniguccio: The Classic Sixteenth-Century Treatise on Metals and Metallurgy, eds. and trans. Cyril Stanley Smith and Marta Teach Gnudi. New York: Dover Publications.

Boardman, John. 1988. Trade in Greek Decorated Pottery. Oxford Journal of Archaeology 7: 27–33.

Boyle, Robert. 1661. The Sceptical Chymist or Chymico-Physical Doubts and Paradoxes, Touching the Spagyrists Principles Commonly call'd Hypostatical: As They Are Wont To Be Propos'd and Defended by the Generality of Alchymists. Whereunto is Premis'd Part of Another Discourse Relating To the Same Subject. London: J. Cadwell for Crooke.

Boyle, Robert. 1691. Will of the Hon. Robert Boyle, F.R.S, July 1691. The National Archives,

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
covers, basins, bell-jars, mortars and many other small instruments.11 These were

8On scientific illustration in late sixteenth-century Florence, see Ligozzi, I ritratti; Tongiorgi
Tomasi & Tosi, “Flora e Pomona”; Bassani Pacht et al., Marie de Médicis; and Garfagnini,

Firenze e la Toscana, vol. II. Some of those objects conceived by clock-makers and scientific

instrument builders can be seen in the exhibition catalogue: Acidini, Magnificenza alla Corte.

9 For more information on the differences and similarities between the Galleria dei lavori and the other European court's workshops, see Kieffer, Ferdinando I de Médicis.

10 "Che nessuno pratichi in detta Fonderia eccetto nella prima stanza dove si distribuischono li medicamenti." (ASF, GM 403, ins. 2, fol. 120).

11 The "terraces" mentioned in the inventory don't exist anymore, but we can deduce from the archival sources they were little open rooms made of wood.

followed by a huge distillation laboratory including an ordigno (machine) to distil

acquavite composed of 60 glass balls, a bain-marie, pallets full of glass pots of all

kind and a pierced bench to place the separatoie (separators). The following room

was full of medicine cabinets and the final room was a smithy including another

terrace, equipped with all the tools traditional to this activity (anvil, pincers,

sledgehammers etc.).

The Origins of the Fonderia: The Casino di San Marco

At the same time, another fonderia in Florence witnessed similar activities. This

was the fonderia in the Casino di San Marco, created by Francesco I long before the
one in the Uffizi. The visitors and chroniclers of the time gave famous accounts of

the experiments Francesco I lead there: for example, he found a way to melt rock crystal and to imitate Chinese porcelain; he made false precious stones, fireworks, and explosives. He experimented with new remedies on dying people, and he practiced alchemy. He did not turn his back on the arts of painting, sculpture, illumination or goldsmithing “and around those things he spends quite all the time in a place called Casino, where he has many rooms with masters who do different works and there he keeps his stills.”12 According to the traditional historiography, the young prince Francesco untiringly frequented Cosimo I’s fonderie, set up in the Palazzo Vecchio, and so recognised in himself the passion that was to prompt him his life long.13 In fact,

as early as 1570, he ordered the architect Bernardo Buontalenti to draw up the plans

for the Casino di San Marco: he intended to transform it into a palace worthy of a

prince and to display there his patronage in arts and sciences. In 1574, when Cosimo

I died, Francesco took possession of the Casino and of its gardens and set up the first
court workshops. Besides the artistic workshops, the palace included a fonderia
well known for its alchemical research, and its unusual layout. Indeed the suite of

adjoining rooms forms a closed circle, a labyrinth that follows a strict structural

logic and hermetic aesthetic similar to that of the Studiolo in the Palazzo Vecchio.14

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Fig. 6 The Casino di San Marco (Courtesy of Museo Galileo, Florence)

36 Covoni, Buontalenti ai tempi Medicei, 12.

Material and Temporal Powers at the Casino di San Marco (1574–1621) 141

many times that it has progressively lost its original design. As far as I know, no

sixteenth-century drawings or engravings of the original disposition of the residen-

tial rooms and the workshops survive today. Pierfilippo Covoni found a plan of the

first floor of the building post-dating the death of Francesco’s son Don Antonio de’

Medici (1576–1621). Since we know that Don Antonio restructured the building in

1594 we should be cautious about putting too much weight on this document

(Fig. 9). At the beginning of the seventeenth century the fonderia occupied a row

of first floor rooms; in Don Antonio’s day, these were preceded by a library, to

which I will return when describing Don Antonio’s activities. We certainly know

that in Francesco’s Casino there existed a glass works, an alchemical laboratory and

a furnace for producing porcelain: painters, goldsmiths and gem carvers were also

active in the Casino but it is not clear where exactly their workshops were sited, and

Fig. 8 Detail of Buontalenti’s window of the Casino di San Marco

Fig. 9 Plan of the first floor of the Casino di San Marco published by Covoni in 1892b

how they related one to another. On the basis of the evidence available it seems that

Buontalenti and Francesco carefully planned the disposition of the arts, putting the

chemical ones at the top of their concerns. In addition to these workshops,

Francesco displayed some of his spectacular collections of paintings, sculptures,

coins, gems, naturalia and mirabilia in several of the Casino’s rooms and the

fountain with Giambologna's sculpture in its garden. If the arrangement of their display is not precisely known, we have several contemporary descriptions which help to provide us with a relatively good picture of what was going on within the walls of the Casino. For example, we have several documents reporting on the salaries of the artists and alchemists employed by Francesco at the Casino.37 In addition to Buontalenti, who coordinated the activities of the various laboratories, the salaried artisans of the Casino in 1580 included the following 18:

'Tanai de' Medici proveditore de' tapezieri scudi 4 [tapestry]

Maestro Antonio portoghese tappeziere, scudi 10 [tapestry]

Ieremia Foresti fonditore, scudi 12 [metal worker]

Niccolò di mastro Sisto, scudi 5 [Medici ‘porcelain’, glass, alchemy]

Filippo della Serena mastro de vetri, scudi 12 [glass]

Giovanni Ambrogio milanese, scudi 20 [silver and rock crystal]

Stefano milanese, scudi 20 [silver and rock crystal]

Giuseppe che lavora con li duoi milanesi, scudi 6 [silver and rock crystal]

Pier Maria detto il Faenzino, scudi 10 [Medici ‘porcelain’ and majolica]

Giuseppo da Campo stovigliaio, scudi 7 [Medici ‘porcelain’ and majolica]

Giuseppe Marchesi vineziano, scudi 30 [ruby maker and carver]

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Zumbulyadis, Nicholas. 2010. Böttger’s Eureka!: New Insights into the European Reinvention of

Porcelain. Bulletin for the History of Chemistry 35: 24–32.

200 M. Wesley

Contents
Introduction
Laboratories and Workshops
Art Technologies and Knowledge of Material Transformation
Epistemic Changes Between Artisans and

Alchemists
Acknowledgements
Bibliography

About the Authors
The Alchemical Art of Dyeing: The Fourfold Division of Alchemy and the Enochian Tradition
Alchemy and the Making of Gold: An Overview of the Byzantine Tradition
Two Early Alchemical Treatises: Ps.-Democritus and Isis
Towards a Definition of Alchemy: Zosimus and the Enochian Myth
Conclusion
Appendix
Bibliography

Transmission of Alchemical and Artistic Knowledge in German Mediaeval and Premodern Recipe Books
Art and Alchemy Within Recipe Books
Mediaeval and Premodern Recipe Books
The Sources and the Context of Production
The Modalities of Composition
The Function(s) of Recipe Books
Reliability of Recipe Books

Alchemical and Art-Technological Recipes Within a Manuscript: Location, Relationship and Distinction
Conclusion
Appendix: List of Manuscripts
Augsbourg, Staatsbibliothek
Bad Berleburg, Schlossbibliothek Sayn-Wittgenstein
Bamberg, Staatsbibliothek
Berlin, Staatsbibliothek
Budapest, Nationalbibliothek
Erfurt, Bibliothek der Stadt
Heidelberg, Universitätsbibliothek
Karlsruhe, Badische Landesbibliothek
Munchen, Bayerische Staatsbibliothek
Nuremberg, Germanische Nationalmuseum
Prague, Narodni Knihovna
St Gallen, Kantonsbibliothek
Trier, Stadtbibliothek
Vaticano, Biblioteca Apostolica Vaticana
Vienna, Österreichische Nationalbibliothek
Winterthur, Stadtbibliothek
Wolfenbuttel, Herzog- August Bibliothek
Zürich, Stadtbibliothek
Bibliography

Artisanal Processes and Epistemological Debate in the Works of Leonardo Da Vinci and Vannoccio Biringuccio
Chemical´ Arts During the Renaissance
Leonardo and the Chemical´ Arts
Biringuccio´s Pirotechnia and the Knowledge of Nature
Bibliography
Artificial Interventions in the Natural Form of Things: Shared Metallogenetical Concepts of Goldsmiths and Alchemists
Metallogenetical Concepts
The Doppelscheuer
Conclusion
Bibliography
The Laboratories of Art and Alchemy at the Uffizi Gallery in Renaissance Florence: Some Material Aspects
The Fonderia in the Uffizi: Origins and Material Organisation
Introduction to the Fonderia
The Origins of the Fonderia: The Casino di San Marco
Similarities and Differences Between the Fonderie

The Activities in the Fonderia: Between Art and Science
Glassmaking
The Forge
The Confectionary
Alchemy
Bibliography
Material and Temporal Powers at the Casino di San Marco (1574-1621)
At the Origin of a Myth
Mapping the Florentine Arts
The Casino di San Marco
Bibliography

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
workplace of the alchemist. All workplaces in which “chemical” operations were

performed were known as laboratories, and as such, artisanal workshops with

furnaces and fire in which “chemical” operations were performed were also

known as laboratories. For example, the places in which sixteenth-century pro-

ducers of fireworks developed their materials were called laboratories.9 In terms of

material culture the laboratorium, for example, one designed and equipped by Count Wolfgang II von Hohenlohe-Weidersheim (1546–1610) in the early seven-teenth century, was remarkably similar to the workplaces of apothecaries, metal-workers and glassmakers.10

Before the emergence of mercantilist states, Renaissance courts were the most important agents in establishing laboratories. The laboratories of Rudolf II (1552–1612) in Prague and Landgrave Moritz von Hessen-Kassel (1572–1632) were thought to bring material and intellectual rewards.11 Moritz even founded a chair of chymiatria at the University of Marburg in 1609.12 The teaching of Johannes Hartmann (1561–1631), who was appointed to the chair, included work in the

laboratory, likewise established at the university. Courts also developed workplaces

in which the work of artisans could be tested. Most famously, in the 1670s, Johann

Joachim Becher (1635–1682) proposed the establishment of a Kunst- und Werckhaus at the court in Vienna.13 It would have included several different sorts

of manufactures: porcelain making, silk and wool weaving, the production of

medicines and glassmaking. Chemical laboratories were to form the core of the

8Klein, “Laboratory Challenge,” and “Apothecary Shops.” For the dissociation of experimental

philosophy from the alchemical laboratory, see Shapin, “House of Experiment.”

9Werrett, Fireworks, 29–30.

10Weyer, Graf Wolfgang II. von

Hohenlohe. For material culture of laboratories, see also Osten,

Alchemistenlaboratorium Oberstockstall; Soukup &Mayer, Alchemistisches Gold, and the shorter
report, Soukop, Osten & Mayer, “Alembics, Cucurbits, Phials,” 25; Howard, La Bibliothèque et le laboratoire; Pereira, “Utili segreti”; Rouaze, “Atelier de distillation”; and Joly, “Laboratoire

alchimique.”
11 Among the numerous publications on court alchemy, see two foundational studies: Moran,

The Alchemical World; and Evans, Rudolf II.
12 Ganzenmüller, “Das chemische Laboratorium.”
13 Smith, Business of Alchemy, 190–8.

house. These laboratories were intended to specialize in salts and acids used in the

production of mineral dyes. Becher envisioned that the manufacture and export of these dyes would finance the other operations of the Kunst- und Werckhaus. He also envisaged the house attracting artisans from various crafts and that it would function as a deposit of recipes and descriptions of techniques. Trained artisans would then be sent out to distribute new processes, new manufactures and inventions. Becher used alchemy to intrigue his patron, who was always interested in the wealth promised by metallic transmutation, and to link the worlds of the court and commerce.

Such court projects often took inspiration from the two earliest examples of spaces bringing together alchemy and the decorative arts at the Medici court in

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
di valuta, e di artifizio grandissimo. Il disegno di questo palazzo è di Bernardo Buontalenti, huomo

di peregrino ingegno e raro, come si vede nelle finestre, che sono leggiadre, nelle camere, che sono

artifiziose, ma quella che è principale in su la strada è bella a maraviglia.” On the Casino, see Covoni, Buontalenti ai tempi Medicei; and Fara, Bernardo Buontalenti, 156–65.

Marco, a complex built in 1442 by the architect Michelozzo Michelozzi (1396–1472) under the patronage of Cosimo il Vecchio (1389–1464), the Casino encompassed the Orti medicei, a symbolic place commissioned by Lorenzo il Magnifico (1449–1492) to host the excellence of Florence's fine arts (Fig. 7). As early as 1574, when Francesco I inherited the Orti from his father he used the existing building as an officina di esperimenti chimici e fisici, thus introducing a

Fig. 5 Francesco I (right) working with his father's crown. Alessandro Fei (Courtesy of Museo di Palazzo Vecchio, Florence)

new hierarchy of the arts.36 The Casino incorporated the old buildings into a

new

and unusual one, which is particularly evident in the bizarre architectural decora-

tion of the windows (Fig. 8). The building, now a tribunal, has been restructured so

Fig. 7 The Casino di San Marco and the Orti Medicei in a detail of Buonsignori's Nova pulcherrimae civitatis Florentiae topographia accuratissime delineata (1584). Private collection

Fig. 6 The Casino di San Marco (Courtesy of Museo Galileo, Florence)

36 Covoni, Buontalenti ai tempi Medicei, 12.

many times that it has progressively lost its original design. As far as I know, no

sixteenth-century drawings or engravings of the original disposition of the residen-

tial rooms and the workshops survive today. Pierfilippo Covoni found a plan of the

first floor of the building post-dating the death of Francesco’s son Don Antonio de’

Medici (1576–1621). Since we know that Don Antonio restructured the building in

1594 we should be cautious about putting too much weight on this document

(Fig. 9). At the beginning of the seventeenth century the fonderia occupied a row

of first floor rooms; in Don Antonio’s day, these were preceded by a library, to

which I will return when describing Don Antonio’s activities. We certainly know

that in Francesco’s Casino there existed a glass works, an alchemical laboratory and

a furnace for producing porcelain: painters, goldsmiths and gem carvers were also

active in the Casino but it is not clear where exactly their workshops were sited, and

Fig. 8 Detail of Buontalenti’s window of the Casino di San Marco

Fig. 9 Plan of the first floor of the Casino di San Marco published by Covoni in 1892b

how they related one to another. On the basis of the evidence available it seems that

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:

Paolozzi Strozzi, Beatrice, and Dimitrios Zikos. 2006. Giambologna, gli dei, gli eroi. Florence: Giunti.

Pelli Bencivenni, Giuseppe. 1779. Saggio istorico della Real Galleria di Firenze, vol. 2. Florence: Per Gaet. Cambiagi stamp. Granducale.

Perifano, Alfredo. 1997. L’Alchimie à la Cour de Côme Ier de Médicis: savoirs, culture et politique. Paris: Honoré Champion.

Pieraccini, Gaetano. 1924–1925. La Stirpe de'Medici di Cafaggiolo: saggio di ricerche sulla trasmissione ereditaria dei caratteri biologici, vol. 2. Florence: Vallecchi.

Pozzi, Mario. 2004. Filippo Pigafetta consigliere del principe. Vicenza: Biblioteca Civica

Bertoliana.

Schepelern, H.D. 1985. Natural Philosophers and Princely Collectors: Worm, Paludanus and the

Gottorp and Copenhagen Collections. In The Origins of Museums: The Cabinets of Curiosities in Sixteenth and Seventeenth-Century Europe, eds. Oliver Impey and Arthur MacGregor, 121–

127. Oxford: Clarendon.

Somers Cocks, Anna. 1980. Princely Magnificence: Court Jewels of the Renaissance, 1500–1630. London: Debrett's Peerage.

Targioni Tozzetti, Giovanni. 1769–1779. Selve di notizie, spettanti all'origine

de’ progressi e
miglioramenti delle scienze fisiche in
Toscani, messe insieme dal Dottor
Giovanni Targioni-
Tozzetti, per uso del dottor Ottaviano
suo figlio, 17 vols. Biblioteca
Nazionale Centrale di

Firenze, Targioni. Tozzetti, Ms. 189.

Tongiorgi Tomasi, Lucia, and Alessandro
Tosi. 1990. “Flora e Pomona”:
L’orticoltura nei
disegni e nelle incisioni dei secoli
XVI–XIX. Florence: Olschki.

Wazbinsky, Zygmunt. 1994. Il Cardinale
Francesco Maria Del Monte, 1549–1626, 2
vols.

Florence: Olschki.

Zobi, Antonio. 1841. Notizie storiche
riguardanti l’Imperiale e Reale
Stabilimento dei lavori di
commesso in pietre dure di Firenze.
Florence: Tip. di Felice Le Monnier.

Zorzi, Elvira Garbero, and Mario
Sperenzi. 2001. Teatro e spettacolo

nella Firenze dei Medic.
Florence: Olschki.

The Laboratories of Art and Alchemy at the Uffizi Gallery in Renaissance. . . 127

Material and Temporal Powers at the Casino

di San Marco (1574–1621)

Marco Beretta

Abstract Built in 1574 by court engineer and architect Bernardo Buontalenti for

Francesco I de Medici, the Casino di San Marco represents a unique example of a

late Renaissance site of alchemical research, art collecting and political court.

Francesco I’s program to enhance the chemical arts and make it into a body of

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Art Technologies and Knowledge of Material Transformation

Laboratories were not only the workplaces of transmutational alchemists. TheUffizi
and the Casino had little in common with the workplaces of alleged goldmakers

destined to fail and bring their families to financial ruin, as famously depicted by

Pieter Brueghel the Elder (c.1525–1569) (Fig. 2). Just as the spaces known as

laboratories encompassed the workplaces of artisans laboring at furnaces and

using materials and chemical procedures to make both everyday items and objects

of the visual and decorative arts, the activity known as alchemy encompassed more

than attempts to make gold.14 Transmutational alchemy was about the trans-

mutation of all base metals into more noble ones, but chrysopoeia was only one

aspect of alchemy. Alchemy also touched on medicine and chemical manufacture.

It was about the chemical production of things—medicines, porcelain, dyes, and

other products as well as the precious metals—and about the knowledge of how to

produce them. In this sense, “art technologies”—materials and techniques to make

art and knowledge of these materials and techniques—overlapped with alchemy.

Alchemy has deep roots in writings on material transformation from Antiquity.

The productive knowledge associated with material transformation was written

down in recipe books. The Leiden and Stockholm Papyri date to the third century

AD. They contain recipes for the making of gold and silver, for the imitation of

precious stones, and for textile dyes. Whether a recipe is about coloring silver to

make it look like gold or the making of an imitation ruby or another artificial

precious stone, all of the recipes in the Papyri are, indeed, about material trans-

formation and color change as related to the manufacture of luxury goods. As

Matteo Martelli shows in this volume, relatively early on a historical process of

selection, appropriation and differentiation resulted in a more

limited definition of

Fig. 2 Pieter Brueghel the Elder, The Alchemist’s Family: Al-gemist, 1558 (Courtesy of the

Kupferstichkabinett, Staatliche Museen zu Berlin)

14 For the scope of early modern alchemy and the notion of ‘chymistry’, see Principe, Secrets of
Alchemy; and Newman & Principe, “Alchemy vs. Chemistry.”

alchemy primarily focused on the making of gold and silver. Already around

300 AD, a distinction was introduced between a limited definition of alchemy as

metallic transmutation and a more encompassing definition including productive

knowledge and various artisanal technologies.

Transmitted to Europe, and translated into Latin and the vernaculars, several

of

the recipes in the Papyri are still found in collections of recipes in the fifteenth and

sixteenth centuries. More importantly, as Martelli remarks in his essay, the scope of

a recipe collection such as the Mappae clavicula, compiled between the ninth and

the twelfth centuries, is as encompassing as that of the Papyri despite the earlier

attempts to limit alchemy to metallic transmutation only. It should not come as a

surprise then that Sylvie Neven finds it difficult to demarcate between alchemical

and art technological recipes in late medieval collections of recipes. Her contri-

bution to this volume shows that alchemical and art technological recipes shared a

concern with the same materials and artisanal processes. The focus on chromatic

transformation already found in the Papyri continues in fifteenth-century recipes.

Although related to laboratory practices, Neven emphasizes that these recipe

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
spaces bringing together alchemy and the decorative arts at the Medici court in Florence. Two essays in this volume deal with these Florentine workplaces: Fanny Kieffer discusses the Uffizi, and Marco Beretta the Casino di San Marco. Founded in 1586 by Francesco I de’ Medici (1541–1587), and developed by his successor, Ferdinando I (1549–1609), the Uffizi housed the fonderia, a workspace for the arts
of fire. Here medicines were prepared

and metals fused. The fonderia was simul-

taneously a pharmacy, an alchemical laboratory, a smithy, a goldsmith’s workshop,

and a confectionary. The activities performed in theUffizi included everything from
preparing jams and sugar sculptures to the production of glass and fireworks (and

less peaceful applications of saltpeter). Francesco I also established the Casino di
San Marco. It included an alchemical laboratory, a furnace for producing porcelain,

and a glass workshop in which, under the direction of Niccolo Sisti, cristallo (so named because it was as clear as rock crystal) was made, as well as all sorts

of colored glass in imitation of precious stones. Other important activities in the

Casino were the fusion of rock crystal, the counterfeiting of precious stones, the production of fireworks, and the preparation of pharmaceutical remedies. Similar activities were developed in the Uffizi and the Casino, and recipes and personnel travelled between the two places. The only exception was Medici porcelain, which was exclusively produced in the workshops of the Casino.

In sum, Renaissance courts established spaces where artisanal workshops and laboratories were brought together facilitating the circulation of materials, people and knowledge between the worlds of craft (today's decorative arts) and alchemy.

Art Technologies and Knowledge of Material Transformation

Laboratories were not only the workplaces of transmutational alchemists. TheUffizi and the Casino had little in common with the workplaces of alleged goldmakers destined to fail and bring their families to financial ruin, as famously depicted by Pieter Brueghel the Elder (c.1525–1569) (Fig. 2). Just as the spaces known as laboratories encompassed the workplaces of artisans laboring at furnaces and

using materials and chemical procedures to make both everyday items and objects of the visual and decorative arts, the activity known as alchemy encompassed more than attempts to make gold.14 Transmutational alchemy was about the trans-mutation of all base metals into more noble ones, but chrysopoeia was only one

aspect of alchemy. Alchemy also touched on medicine and chemical manufacture.

It was about the chemical production of things—medicines, porcelain, dyes, and

other products as well as the precious metals—and about the knowledge of how to

produce them. In this sense, “art technologies”—materials and techniques to make

art and knowledge of these materials and techniques—overlapped with alchemy.

Alchemy has deep roots in writings on material transformation from Antiquity.

The productive knowledge associated with material transformation was written

down in recipe books. The Leiden and Stockholm Papyri date to the third century

AD. They contain recipes for the making of gold and silver, for the imitation of

precious stones, and for textile dyes. Whether a recipe is about coloring silver to

make it look like gold or the making of an imitation ruby or another artificial precious stone, all of the recipes in the Papyri are, indeed, about material trans-formation and color change as related to the manufacture of luxury goods. As

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
from the Apparato della Fonderia di Don Antonio recommended using syrups, fruit

pastes, marmalades or candies to conserve the properties of the active ingredients or

amplify their effects. Osio was also in charge of making sugar medicines containing

plants and distillations from the alchemical laboratory.42

Alchemy

The most important and best-documented activity was by far alchemy

(or chymistry) with its various practical applications. The production had several

forms, according to the needs of the court: perfumes and cosmetics, remedies of all

kinds, poisons and antidotes. In contrast to the production of the Casino which, as

we saw, was well known due to the writing of treatises and recipes books, the

fonderia, as far as we know, did not produce any written documentation. So it is

only possible to assess the situation through expense forms, orders and letters of

instruction to Sisti, the maestro della Fonderia, and, above all, through the dosage

booklets given with the remedies.

39 On the confectionary laboratory, see Kieffer, “Confiserie des Offices.”
40 ASF, GM 124, fol. 173; and GM 183, ins. 21, fol. 29.
41 ASF, GM 124, fol. 209’; and GM 183, ins. 2, fol. 21.
42 On sugar and medicine, see Kieffer, “Confiserie des Offices.”

The dosage booklets are a distinguishing feature of the fonderia of the Uffizi. They demonstrate that Uffizi production was designed to be distributed either as gifts, or as sales products. While the Casino concentrated on experimentation and methodological or cultural—even mystical—thought, in the Uffizi the recipes have very little variation and the packaging as well as the dosage booklets are mass-produced. For the more famous consignees (such as Cardinal Gioiosa) these dosage

booklets are illustrated and sometimes even gilded.

To meet occasional surges in demand, Sisti had to insure a steady supply of raw

materials. Sometimes he had to request Cosimo Latini, the minister of the Gallery,

to obtain rare or very expensive substances, such as amber or musk.43 However,

most of the raw material came from the Vallombrosa abbey. The monks there grew

medicinal plants for the Uffizi and collaborated directly with the fonderia, as was
the case for someone called Giovanni di Giuliano da Montereggi, mentioned as a

erbolaio or herbolista (herbalist) with his assistant Marco di Simone. Both were

paid on a daily and merchandise basis, and don’t appear in the court roles.44

The supply bills coming from Vallombrosa also show which plants are used in

the Uffizi and for which purpose.45 We realise that the plant ingredients of certain

medicines for the grand duke correspond exactly to the recipes recorded in the

Apparato della Fonderia di Don Antonio, so that our hypothesis of a collaboration

between both fonderie during Ferdinando I’s reign was reinforced. Here, for

example, is a supply bill for herbs from the archives:

The day 25th of October, in Florence. The Lord Cosimo Latini director of the Gallery. His

Grace would be pleased to pay [. . .] to have brought to His Grace’s Fonderia juniper berries
to make the oil for the Petechiae Water and to have served for two days mashing the berries

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Omnifaria Doctrina) and some specific essays (see Bidez, Épitre sur la Chrysopée, 51–70; and
Duffy, Michaelis Pselli, texts 19–31). On the relations between Psellos’s alchemical interests and

his investigation of the physical world, see Katsiampoura, “Transmutation of Matter,” 665–7.
4 “τὸ μὲν γὰρ ὕδωρ πηγνύμενον λιθοῦται ει$ ς κρύσταλλον [and] τοῦτο ει$ ς ἀτμίδα λυόμενον ἀὴρ καθίσταται.”

5 In § 4 Psellos explains the petrifaction of an oak struck by a lightning strike: a quick and sharp lightning does not only make the oak black, but consumes all its humidity and transforms the wood into stone.

6 See infra, § 3.

7 The entry continues by telling how the emperor Diocletian (284–305) made to burn all the Egyptian books on alchemy (in the Greek text: “περὶ χημείας χρυσοῦ και ἀργύρου”); this information probably depends on the Chronicle (Ἱστορία χρονική) of John of Antioch (active under Heraclius’s reign, 610–641): see fragment 280 in Roberto, Ioannis Antiocheni Fragmenta.

Empire, the same definition had been already given by the Persian scholar al-Nadīm

(ninth century AD), who devoted the tenth chapter of his Kitāb al-Fihrist (Book of
Catalogue) to the s
.

ināʻat al-kīmīyā’ (art of alchemy): at the very beginning of the

chapter this art is defined again as “the making of gold and silver from other

metals.”8 More than one century later, the Syriac scholar Bar Bahlul (eleventh

century AD) dealt with the same subject in two entries of his lexicon:

1. Duval, Lexicon Syriacum, vol. I, 901: Kimiya, the dark stone. It is the work of the art of gold and silver. And there is

someone who explains this word from the name kima (the constellation of the

Pleiades), that is the height stars, because (this art) is accomplished by means of

the height mixtures.9

2. Duval, Lexicon Syriacum, vol. I, 904: Kemela’a, that is the alchemy (kimiya’a) of the seven bodies and of the various natures; [it includes] arts such as goldsmith’s art and copper, iron, and glass working.10

Only the last definition includes a wide set of techniques applied to different substances (among which glass is mentioned), while most of the above-quoted texts identifies alchemy with the making of gold and silver. The two precious metals were probably the most attractive outcomes that anyone engaged in alchemical

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
drian Greek papyri preserved in Leyden and Stockholm.7 These date from the third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

They also have recipes dedicated to the art of dyeing, to chrysography, and to the

making of artificial precious stones.

An echo of these recipes can be found in the Codex Lucensis 490 (Lucca,

Biblioteca Capitolare Felinià na), also known as the Compositiones ad tingenda musiva or Compositiones variae. This manuscript, dated from the end of the eighth

or the very beginning of the ninth century, is based on previous Greek sources

compiled around the beginning of the seventh century.9 This recipe book deals with

various artistic techniques, notably the dyeing of skins, the manufacture of pig-

ments, colorants, varnishes and glues, chrysography and the gilding of metals.

The Lucca Manuscript shares content with the Mappae clavicula, compiled

around 800. Parts of this text have far earlier origins and present parallels not

only with the Leyden and Stockholm papyri but also with Syriac translation of

Zosimus and with ps.–Democritus's writings.10 The nucleus of this text was

probably a Greek alchemical treatise written and translated between the fourth

and the fifth centuries, with additions in the eleventh and twelfth centuries.11 The

two main manuscripts are the Corning, Corning Glass Museum, Ms. 5 and Sélestat,

Bibliothèque Humaniste, Ms. 17, but this tradition was also (partly) disseminated

through dozens of manuscripts. The Mappae clavicula contains about 300 recipes

6 Singer, Catalogue of Latin, 38; Principe & DeWitt, Transmutations, 2–3; and Principe, Secrets of Alchemy, 13.

7 On Leyden, see Rijksmuseum van

Oudheden, Papyrus P. LEID.X.; On
Uppsala, Victoria

Museum, P. HOLM.
8Halleux, Papyrus de Leyde.
9 Hedfors, Compositiones ad tingenda
musiva; and Johnson, Compositiones
variae.
10 Berthelot & Ruelle, Anciens
alchimistes grecs. See also Martelli,
Pseudo–Democrito.
11 Halleux & Meyvaert, “Mappae
clavicula”; and Berthelot & Duval,
Chimie au Moyen-Âge,
vol. I.

and descriptions of miscellaneous
chemical operations, including
instructions for

the manufacture of dyes and pigments,
for the gilding and painting on glass,
as well

as, among others, metalwork,
chrysography, distilling alcohol, making

candy, and

creating military devices.12

Mediaeval and Premodern Recipe Books

In mediaeval and premodern times, artistic and alchemical procedures were often

described within compilations of texts that may concurrently address various fields

such as medicine, cooking, botany or pharmacology. They also include magical

recipes, dietetical instructions or advice on home-economics. All these various

disciplines are embedded within the genre of the Fachliteratur.13 This kind of

literature regroups all texts of a utilitarian and informative nature whose content

does not principally concern aesthetic or religious issues, or matters relating to

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
several (re)interpretations since the eighteenth century, see Principe & Newman, “Historiography

of Alchemy.”
4Halleux, “Alchimie,” 336–7; Newman, “Technology and Alchemical Debate,” 432–3; Pereira,

“Use of Vernacular Languages,” 336; and Kahn, Alchimie et paracelsisme, 7–8.
5 Halleux, Savoir de la main, 134.

24 S. Neven

of base or impure matter (often metal or stone) into a noble or perfect one.6 To do

so, alchemists used to perform chemical processes and manipulations which resem-

bled those practiced by contemporary artists and artisans.

Thus, in both fields particular importance is placed on craft practices. Both

alchemical and artistic recipe books describe various processes for purifying and

transforming materials, either for improving their properties or in order to use them

for specific purposes. In this context, from a technological point of view, the term

‘transmutation’ could refer to the colouring of glass, the melting and tinting of

metals, the dyeing of stones or gems, or the manufacture of synthetic pigments. A

huge number of recipes are dedicated to procedures for obtaining gold or silver or

gold-silver like substances from base metals (copper, tin, lead, iron, mercury).

The tradition of recipe books has roots deep in Antiquity. Treatises dealing both

with art-technological and alchemical procedures notably survived in the Alexan-

drian Greek papyri preserved in Leyden and Stockholm.7 These date from the third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

They also have recipes dedicated to the art of dyeing, to chrysography, and to the

making of artificial precious stones.

An echo of these recipes can be found in the Codex Lucensis 490 (Lucca,

Biblioteca Capitolare Felinià na), also known as the Compositiones ad tingenda musiva or Compositiones variae. This manuscript, dated from the end of the eighth

or the very beginning of the ninth century, is based on previous Greek sources

compiled around the beginning of the seventh century.9 This recipe book deals with

various artistic techniques, notably the dyeing of skins, the manufacture of pig-

ments, colorants, varnishes and glues, chrysography and the gilding of metals.

The Lucca Manuscript shares content with the Mappae clavicula, compiled

around 800. Parts of this text have far earlier origins and present parallels not

only with the Leyden and Stockholm papyri but also with Syriac translation of

Zosimus and with ps.–Democritus’s writings.10 The nucleus of this text was probably a Greek alchemical treatise written and translated between the fourth and the fifth centuries, with additions in the eleventh and twelfth centuries.11 The two main manuscripts are the Corning, Corning Glass Museum, Ms. 5 and Sélestat, Bibliothèque Humaniste, Ms. 17, but this tradition was also (partly) disseminated through dozens of manuscripts. The Mappae clavicula contains about 300 recipes

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
literature quoted there.
16 Newman, Promethean Ambitions, 115–63.
17 Principe & DeWitt, Transmutations, 11–27.
18 For experts and entrepreneurial alchemy, see Nummedal, Alchemy and Authority, 40–5.

It is in this context of alchemical expertise that we should consider the artisans

who adopted the language of alchemy in the early modern period to attract the

attention of patrons. One example can be seen in the description of Black Berthold,

the legendary discoverer of gunpowder in Europe, in the late fifteenth-century

manuscript Feuerwerckbuch. Depicted in the company of a furnace and alembics,

Berthold’s creation of ordnance is identified with alchemy. Black Berthold, the

Feuerwerckbuch reads,

worked with the great alchemy like those masters who are engaged with precious and

valuable things, with silver and gold, and with similar metals. These masters can separate

silver and gold from other precious jewelry, and from other valuable colors which they can

produce. Now this master Berthold wanted to induce a golden coloration. For this

he used

salpetre, sulphur, lead and oil. Then he put these ingredients in a container made of copper,

which he sealed completely, exactly as it should be done, but when he put it on fire and the

container became hot, it burst into many pieces.19

The Feuerwerckbuch witnesses material production based on practical chemistry in

terms borrowed from alchemical transmutation, or more precisely, the making of

gold. The analogy was not always with chrysopoeia as such. Vasari’s already

mentioned fashioning of Van Eyck as an alchemist shows that the image of alchemy

as a field of expert inquiry on materials and material transformation was considered

sufficiently positive to identify with. However, we should not forget that Vasari's

identification of Van Eyck as an alchemist served the purpose of downplaying

Netherlandish art as techne, only a first step in a narrative of art historical progress

culminating in Vasari's beloved Florentine art.

The epistemic value of techne was shifting at the time of Vasari. Artisans came

to be considered experts of nature likening the artisanal processes of material

transformation undertaken in their workshops to those of nature.20 As Andrea

Bernardoni shows in this volume, Biringuccio's Pirotechnia is part of this larger

process of shifting epistemic value of artisanal processes. Rejecting

transmutational

alchemy as “false” and the alchemists who practiced it as fraudulent, Biringuccio

carved out space for true alchemy as one of the arts of fire. Artisanal workshops,

such as Andrea Verrocchio’s (c.1435–1488) in which Leonardo da Vinci (1452–

1519) apprenticed or Leonardo’s own workplace, shared a material culture with

alchemical laboratories. As we have already pointed out, artisans used similar

equipment and performed “chemical” operations. However, Bernardoni argues,

Biringuccio made the claim that these artisans were the true experts on matter,

materials and material transformation and that artisanal “chemical” operations were

the key to natural knowledge.

One of the readers of Biringuccio’s Pirotechnia was Johannes Mathesius (1504–1565), a Lutheran preacher in St. Joachimsthal, the center of an important mining district. In her contribution to this volume, Henrike Haug analyzes Mathesius’s

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
16 The collaboration between Leon Battista Alberti and Filippo Brunelleschi opened a new setting

for the social relations between the world of learning and the creative craftsmen. One century later

a literary figure such as Benedetto Varchi could take special pride in boasting of his friendships

with craftsmen.
17 “Fiorenza è citta fondata sull’arti e traffichi mercantili,” cited in

Corazzini, Diario Fiorentino, 273.

18 Baroni Vannucci, Jan Van Der Straet, 397–400.

reflected the grand duke's alchemical interests.19 This emblematic space had been preceded by Cosimo I's fonderia, a distillery located among a few court workshops and the many display rooms of the Palazzo Vecchio, and would be followed by the network of workshops and laboratories at the Casino di San Marco, the fonderie of the Uffizi Gallery and Palazzo Pitti.20 Throughout the city innovative spaces with new purposes were created during this period: natural history museums annexed to botanical gardens filled with rare plants, the ateliers of glassmakers, and the

creation of new and more specialized apothecary shops, while traditional institu-

tions such as hospitals and public squares assumed new forms. Even such familiar

landmarks as the Loggia dei Lanzi were invested with new significance during the

Cinquecento; originally the site of public assemblies and ceremonies, the Loggia

was now even used on occasion to display natural history exhibits from the Medici

collections for the edification of the general public and, housed a botanical garden

on the roof.21 Monographs on some of these sites have been published, most of

Fig. 2 First engraving and Luigi Alamanni’s and Stradano’s Nova Reperta (1587–1589) (Cour-

tesy Museo Galileo Florence)

19 Although Stradano’s representations of the arts did not explicitly refer to Florence in the

captions, several visual details reveal their connection with the Tuscan capital.

20 On the history of the fonderie, see Piccardi, “Fonderia Medicea di Firenze.”

21 Vossilla, “Cosimo I, lo scrittoio del Bachiacca.”

them by art historians who have underlined the role of the Medici as patrons of the

arts.22 Important as their work has been in documenting the historic relevance of the

new architectural settings, a global picture is still lacking of the nexus between

these sites dedicated to the production of scientific and technical knowledge and the

economic forces that propelled their realization. A recent book on the

history of the

Spezieria al Giglio illustrates the profound economic and cultural changes intro-

duced by apothecaries who were only apparently immobile.23 Cosimo I’s policy of

sponsoring the arts and technology as part of his program of economic expansion

forced intellectual elites to take account of a body of knowledge that had been

confined to the margin of the Court, and to re-evaluate their room for manoeuver,

not only conceptually but also architectonically, in terms of their work spaces.

The Casino di San Marco

In this process of reconfiguring Florentine arts and crafts, alchemy played an

exceedingly important role, both as a fashionable scientific discipline with

not

particularly strong academic ties and as a useful set of experimental practices in

which several crafts helped artisans quench their thirst for innovation and realize

their socio-cultural ambitions. I shall not explore here the spread of interest in

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
craftsmen and artists as much as it praises the transformative force of fire turning

relatively cheap and humble materials into highly valued objects of art. These

objects were all products of the arts of fire, which according to Vannoccio

Biringuccio (1480–c.1539) included alchemy, and excluded “false alchemy”

concerned with pretentious transmutation.2

In the background of Van Utrecht’s painting, a window opens on to a space in which a man stands working at a stove with an open fire. Stirring a cauldron, the man is shown in the material company of bellows, an anvil, a melting and a distilling furnace, and other equipment related to the worlds of assaying and metallurgy. Van Utrecht’s background refers to the spaces in which the objects in the foreground were produced. Were these spaces laboratories or artisanal work-shops? Were they home to gold- and silversmiths, glassmakers or producers of porcelain? The difference between workshops and laboratories during this period is ambiguous, not only in Van Utrecht’s depiction, and the line between the two

spaces is as difficult to draw as that between the arts of fire and alchemy. This book

is concerned with the interconnections and differentiations between foreground and

1My discussion of Van Utrecht’s painting in this and the next paragraph is based on Göttler,

“The Alchemist, the Painter.”
2 For Biringuccio’s attitude towards alchemy, see Newman, Promethean Ambitions, 128–32.

background in Van Utrecht’s painting, between artisanal workshops and alchemical

laboratories, between the material arts and alchemy.

Van Utrecht is one of the heirs of Van Eyck (c.1390–1441). Following Giorgio

Vasari (1511–1574), Karel van Mander (1548–1606) portrayed Van Eyck as an

alchemist and likened his invention of oil paint to alchemical experimentation.3

According to VanMander, Van Eyck experimentedwith varnishes and binding agents,

“and found after much investigation that pigments mixed with such oils became

malleable and dried hard, and having dried became impermeable, and that the oil

made colours livelier, and that they themselves became lustrous without varnishing.”4

It turns out that Van Eyck did not invent oil paint, and neither was he an alchemist.

Nevertheless he would never have been described in those terms had not artists

and alchemists both been considered agents of material transformation. Van

Mander’s description of Van Eyck’s experimentation also reminds us that painters

and alchemists shared materials, and that painters used artificially created pigments,

“made by alchemy” according to Cennino Cennini (c.1370–c.1440). However, this

Fig. 1 Adriaen van Utrecht, Allegory of Fire, 1636 (Courtesy of Royal Museums of Fine Arts of

Belgium, Brussels (Photo: J. Geleyns))

3 Davis, “Renaissance Inventions.”
4Mander, Het schilder-boeck, 199v. Translation in Melion, Shaping the Netherlandish Canon, 79.

book is not concerned with travelling materials and shared material culture.

Shifting the focus from painting to the decorative arts, this book scrutinizes

epistemic exchanges between producers of the arts of fire and alchemists.

Laboratories and Workshops

What can the evolution of the laboratory, and its shifting relation to the artisanal

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
they also have the same types of production and dissemination? Using a corpus of

about 40 manuscripts produced in Northern Europe between the fourteenth and the

sixteenth centuries, this paper investigates the context behind these writings, and the

various ways alchemical and artisanal recipes were embedded within recipe books.

It also proposes some clues to assist in locating, identifying and demarcating

alchemical writings within the literature of recipes.

In the Middle Ages and premodern period, alchemical knowledge and practice was

frequently transmitted via collections of recipes grouped concomitantly with artistic

instructions. Presented in the form of a succession of more or less short notes, these

S. Neven (*)

FRS-F.N.R.S, University of Liège (ULg), ‘Transitions’ Liège, Belgium

e-mail: Sylvie.Neven@ulg.ac.be

S. Dupré (ed.), Laboratories of Art, Archimedes 37,

DOI 10.1007/978-3-319-05065-2_2, © Springer International Publishing Switzerland 2014

mailto:Sylvie.Neven@ulg.ac.be

writings describe processes for the manufacture, preparation and application of

various types of materials and substances. The majority are anonymous compi-

lations of texts, which may originate from older or undetermined authorities.

Hundreds of such collections of recipes dealing both with alchemical and

art-technological procedures were produced and disseminated in Northern Europe

from the fourteenth century on, especially in German-speaking countries.

Drawing on a delimited corpus of about 40 representative German manuscripts

dated from the fourteenth to the sixteenth century, this paper investigates the

connections and similarities between these two fields and examines the various

ways alchemical and artistic instructions were embedded within recipe books.1 It

argues that textual form and lexical proximities within recipes from these different

disciplines may lead to association, contamination and confusion within this textual

genre. It finally suggests some clues to help locate, distinguish and demarcate

alchemical content within the literature of recipes.

Art and Alchemy Within Recipe Books

At first sight, any overlap between alchemy and art-technology within recipe books

can be broadly explained by the mutual use of various materials and substances

such as “common stones, gems, and types of marble, gold and other metals, sulfurs,

salts, and inks, azures, minium, and other colors, oils and burning pitches,

and

countless other things.”2 More precisely, the field of art-technology encompasses a

large range of craft practices involved in the production of pieces of art (including

those which incorporate such substances). This ‘hand’ knowledge, is related to the

mechanical arts and is divorced from the philosophical or speculative dimension.

Yet, alchemy could be described as the practical, philosophical and medical search

for the perfecting of base material substances and also for the extending of life.3

The theoretical and practical aspects of alchemy involve both the study of all

inanimate or animate things made from the elements and the observation and

imitation of natural processes within the laboratory.4 In this context,

alchemy

could be seen as a mechanical art, in the sense that it works on matter but is also

a liberal art, as it attempts to explain matter in its composition and its transforma-

tion.5 In the practical sense, one of the main goals of alchemy is the transmutation

1 The main data and characteristics of these manuscripts are given in Appendix.
2 Brewer, Fr. Rogeri Bacon, 39–40.
3 Perception and definition of alchemy is not chronologically constant and has been the subject of

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Material and Temporal Powers at the Casino di San Marco (1574–1621) 137

similar to the one illustrated in Biringuccio’s Pirotechnia (fol. 43v), where the

techniques of shaping glass artifacts both by free blowing and by blowing into a

mold are shown. The furnace painted by Butteri suggests that he was illustrating the

glass works set up by the Venetian glassmaker Bortolo who arrived in Florence in

Fig. 4 Francesco I (left) visiting his glass work. Giovanni Maria Butteri (1570–1572) (Courtesy

Museo di Palazzo Vecchio, Florence)

1569 when the Studiolowas taking shape.31 This is confirmed by the fact that on the

right side of the painting, we see material for the construction of a new furnace and

on the left Francesco is examining finished piece of glassware. The size of the glass

works is impressive and the entrance shown in the background of the painting with

a couple of distinguished Florentine courtiers or citizens just inside the door opens

up a glimpse of the Loggiato of the Galleria della Uffizi.32 Between Stradano's

painting of the alchemical laboratory and Butteri's of the glass works, a painting by

Alessandro Fei (1543–1592) depicts a man who seems to be Prince Francesco

working on the gem-studded grand ducal crown in the atelier of a goldsmith and

jeweler (Fig. 5). The arrangement of the painting suggests that glassmaking and

jewelry were the arts most closely connected with alchemy. Their connections

inspired Francesco to create a new decorative scenario, dominated by alchemy,

by which illustrate the relation between arts and nature.

Francesco was extremely keen, even more than his father, to promote the study

of the natural sciences and the arts related to them. His passion for the

chemical arts

was so strong that in 1560, when he was only 19, he was told off by his brother

Giovanni for attending the works of the fonderia all day long.33 It was Francesco

who, in September 1569, charged the architect, engineer and inventor Bernardo

Buontalenti (1531–1608), who had been one of his teachers since 1550, to oversee

the construction of Bortolo’s new glass furnace and a new fonderia.34 Francesco I

also exploited Buontalenti’s versatile skills in the works of the fusion of precious

stones and, in particular of rock crystal. Francesco’s project to enhance his multi-

faceted interest in the alchemical and chemical arts found a new, imposing site with

the construction of the Casino di San Marco (1567–1574), a sumptuous palace

designed by Buontalenti (Fig. 6).35 Across the street from the Convent of San

31As pointed out by Heikamp. Studien, 63–9.

32 At the time the ground floor of the Uffizi was already completed. I thank Suzy Butters for

providing me with this information.

33 “Guardi di non si profonder troppo nel piacer della Fonderia, che qua vien detto, che ella non

esce mai et massimamente il giorno; talché al ritorno nostro speriamo di veder qualche nuova e

Visible: 0% - 95%

ASSISTANT

Great — I found
mentioned in the text that weren't in the first list of 50. Many come from recipe books, Medici laboratories, and artisanal practices like porcelain, confectionery, fireworks, and pharmacy. Here are more to add:

---

51. – Venetian-inspired glass, as transparent as rock crystal 【12:9†source】.
52. – Used to imitate precious stones (emeralds, rubies,

sapphires)【12:9†source】.

53. – Experimental ceramics linked to alchemy【12:1†source】.

54. – Essential to porcelain manufacture.

55. – Another porcelain raw material.

56. – Main ingredient in glass and porcelain【12:1†source】.

57. – Glass flux and soap-making.

58. – Used in fresco, ceramics, and whitening.

---

59. – Textile dye, mentioned alongside Tyrian purple【12:12†source】.

60. – Red dye for textiles.

61. – Yellow dye and medicinal ingredient.

62. – Imported insect dye, important in Renaissance dyeing.

63. – Black/purple dye, early modern use.
64. – Perfume and medicinal substance, expensive court supply 【12:10†source】.
65. – Resin used in perfumes, varnishes, and medicine 【12:10†source】.
66. – Distilled for oils and remedies 【12:10†source】.
67. – Perfume, medicine, and confectionary.
68. – Base in remedies and preservation.

---

69. – In distillation and medicine 【12:2†source】.
70. – For etching metals.
71. – For dissolving gold.
72. – Used to preserve active ingredients 【12:10†source】.

73. – Courtly displays, also medicinal 【12:9†source】.
74. – Prepared in alchemical-pharmaceutical labs 【12:10†source】.
75. – Part of alchemical experimentation in courts 【12:10†source】.
76. – Court remedies against poisons 【12:10†source】.

---

77. – Fuel in forges, glass, and gunpowder.
78. – Saltpeter, sulfur, charcoal 【12:14†source】.
79. – Courtly experiments with fireworks 【12:9†source】.
80. – Containers in recipes and fireworks 【12:14†source】.
81. – Used in casting.

---

82. – Recipes for writing and illumination【12:12†source】.
83. – Artisanal binding agents【12:12†source】.
84. – Recipes for coloring hides【12:12†source】.
85. – Subject of alchemical transformation【12:11†source】.
86. – The recipe texts themselves preserve material knowledge【12:12†source】.

📌 So far we have at least documented. These expand beyond the initial metals, minerals, and pigments to include

— all showing how in courts and workshops.

Do you want me to

with

columns for *Material*, *Alchemy use*, *Artisanal use*? That would give you a clean, at-a-glance resource.

USER

make a list of artistic processes

ASSISTANT

{"queries": ["artistic processes mentioned in the book", "artisanal and artistic techniques in alchemy and art technology", "processes in dyeing, glassmaking, metallurgy, and porcelain", "artistic techniques in the Uffizi and Casino di San Marco", "artistic or artisanal processes in recipe books"]}

TOOL

[Web browsing]

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
transmitted for the purposes of literary preservation, not directly connected with

contemporary workshop or laboratory practices.48

First of all, the textual environment and the diversity of the subjects bound

together with the artistic and alchemical recipes in a same book, lead to the

conclusion that these compilations were mainly read by scholars primarily inter-

ested in natural philosophy and were not intended for contemporary practical use.

Moreover, it has been frequently stated that craft practices were transmitted orally,

Fig. 1 Additional instruction due to Seidel, Munchen, Cgm 4117, fol. 1rv (Courtesy of

Münchener Digitalierungszentrum)

48 Clarke, “Codicological Indicators”; and Neven, Recettes artistiques, 16–23.

from the master to the apprentice.49 A large number of the manuscripts of this study

result from copying and compilation processes undertaken by scribes. As they were

copied in a context outside the workshop or the laboratory, these recipes were not

revised and, consequently, conveyed an anachronistic technical tradition that

became more and more outdated.

Such observations seem to argue against the view that sees these books as

manuals written for the practitioner. But neither were these compendium written

purely for scholarly purposes, deprived of any practical function. In parallel to the

data that could be considered part of the technical heritage of a earlier period, these

recipe books also contain more recent practical instructions—coming from contem-

porary artists and practicing scholars or from the scribe’s own experiments, as the

examples of Seidel, Freising and Schobinger discussed above illustrate. Even when

the writing of these instructions, verbalized in the rhetoric of the recipes, was

carried out by scribes, data were not blindly copied. Scribes organised, assembled,

completed or corrected when they felt it necessary. Thus, even if they were not the

author per se, in the sense that they were not the origin or the source of the technical
or chemical procedures they wrote down, they accomplished a set of activities

linked to ‘authorship’.50 Scribes also made attempts to ensure that the recipes could

be consulted at need: they composed tables of contents or indexes, they introduced

titles within the margins and many other details which attest to a real desire to

deliver usable information. In this context the marginal notes and additions made by

the scribes/authors of the recipe book are of interest as most of them are technical

comments testifying practical interest in both alchemical and artistic instructions.

Several marginal annotations due to Seidel’s hand punctuate the Cgm 4117 and

consist in personal commentaries regarding the technical procedures he records.

For example, on folio 53r, Seidel compares two ways for the melting of crystal.

Concerning the first process he states in the margin that this ‘art’ was not of use to

him as a better (method) is delivered on 219.51 Then on folio 219, he indicates

another method for the same technical procedure, giving as title “How one should

masterfully melt crystal.”52

In this sense, the scribes at the root of these recipe books created not simply a

copy but a unique work, which reflected their own interests, their cultural and life

context and sometimes their intention, which was to deliver practical and useful

instruction.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Art Technologies and Knowledge of Material Transformation

Laboratories were not only the workplaces of transmutational alchemists. TheUffizi
and the Casino had little in common with the workplaces of alleged goldmakers

destined to fail and bring their families to financial ruin, as famously depicted by

Pieter Brueghel the Elder (c.1525–1569) (Fig. 2). Just as the spaces known as

laboratories encompassed the workplaces of artisans laboring at furnaces and

using materials and chemical procedures to make both everyday items and objects

of the visual and decorative arts, the activity known as alchemy encompassed more

than attempts to make gold.14 Transmutational alchemy was about the trans-

mutation of all base metals into more noble ones, but chrysopoeia was only one

aspect of alchemy. Alchemy also touched on medicine and chemical manufacture.

It was about the chemical production of things—medicines, porcelain, dyes, and

other products as well as the precious metals—and about the knowledge of how to

produce them. In this sense, “art technologies”—materials and techniques to make

art and knowledge of these materials and techniques—overlapped with alchemy.

Alchemy has deep roots in writings on material transformation from Antiquity.

The productive knowledge associated with material transformation was written

down in recipe books. The Leiden and Stockholm Papyri date to the third century

AD. They contain recipes for the making of gold and silver, for the imitation of

precious stones, and for textile dyes. Whether a recipe is about coloring silver to

make it look like gold or the making of an imitation ruby or another artificial

precious stone, all of the recipes in the Papyri are, indeed, about material trans-

formation and color change as related to the manufacture of luxury goods. As

Matteo Martelli shows in this volume, relatively early on a historical process of

selection, appropriation and differentiation resulted in a more

limited definition of

Fig. 2 Pieter Brueghel the Elder, The Alchemist’s Family: Al-gemist, 1558 (Courtesy of the

Kupferstichkabinett, Staatliche Museen zu Berlin)

14 For the scope of early modern alchemy and the notion of ‘chymistry’, see Principe, Secrets of
Alchemy; and Newman & Principe, “Alchemy vs. Chemistry.”

alchemy primarily focused on the making of gold and silver. Already around

300 AD, a distinction was introduced between a limited definition of alchemy as

metallic transmutation and a more encompassing definition including productive

knowledge and various artisanal technologies.

Transmitted to Europe, and translated into Latin and the vernaculars, several

of

the recipes in the Papyri are still found in collections of recipes in the fifteenth and

sixteenth centuries. More importantly, as Martelli remarks in his essay, the scope of

a recipe collection such as the Mappae clavicula, compiled between the ninth and

the twelfth centuries, is as encompassing as that of the Papyri despite the earlier

attempts to limit alchemy to metallic transmutation only. It should not come as a

surprise then that Sylvie Neven finds it difficult to demarcate between alchemical

and art technological recipes in late medieval collections of recipes. Her contri-

bution to this volume shows that alchemical and art technological recipes shared a

concern with the same materials and artisanal processes. The focus on chromatic

transformation already found in the Papyri continues in fifteenth-century recipes.

Although related to laboratory practices, Neven emphasizes that these recipe

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
unrelated to it but proper, since it is wholly dependent on the agency and power of fire if it is

to be brought to its perfection. Moreover, the potter’s glazes and colors are all substances of

various metals or impure minerals and therefore belong to fire. Since it is my intention to

treat of fire, minerals, and metals for you, I surely should not have omitted this, particularly

because it is a necessary art which enriches and is greatly praised both for its ingenuity and

its beauty.4

Like the other arts treated in the Pirotechnia, the action of the fire on base earths
and minerals provided integral transformation in the objects shaped by the potter’s

hands. At the time of Biringuccio’s writing, the potter’s art was on the threshold of a

paradigm shift that would create a marked division between craft knowledge and

theory based practice, setting two separate courses for ceramic innovation in the

later sixteenth and seventeenth centuries. His citation of alchemy as one of “two

sources as [pottery’s] principal basis,” is referential to the provision of purified

minerals and “elemental mixtures.”5 However, it also foreshadowed the intellectual

engagement of chymists in the quest for the production of European porcelain,

bringing the weight of the older alchemical tradition to bear.

Prior to the fourteenth century, European pottery was external to the hierarchy of

the esteemed arts, such as metallurgy, glassmaking, painting, and dyeing. The

extensively copied treatise De diversis artibus, composed in the thirteenth century

by the Benedictine monk Theophilus, thoroughly introduces the full range of

artisanal productions valued in Europe in its first book.6 Pottery is entirely absent

from this discussion of laudable arts and only appears once in the text as a canvas

for enameling.7 This is a continuation of pottery's position in the classical period,

when earthen objects were valued as trade goods, everyday wares, or a medium for

other decoration, not for their individual aesthetic form.8

It was not until advancements in the production of luxury quality European

pottery during the fifteenth and sixteenth centuries that the cultural perception of

the material changed significantly, driving engagement by the upper classes and

natural philosophers. An example can be found in a letter from Lorenzo de Medici

(1449–1492) to the Malatesta family in 1490, favorably comparing their gift of

maiolica to silver objects, which is one of the earliest pieces of documentary

4 Biringuccio, Pirotechnia (1942), 392.
5 Biringuccio, Pirotechnia (1942), 392.

6 For various editions of the treatise, see Theophilus, Essay Upon Various Arts (1847), The Various Arts (1961), and On Diverse Arts (1963).

7 “Quam si diligentius perscruteris, illic inuenies quicquid in diuersorum colorum generibus et

mixturis habet Graecia, quicquid in electrorum operositate seu nigelli uarietate nouit Ruscia,

quicquid ductili uel fusili seu interrasili opere distinguit Arabia, quicquid in uasorum diuersitate

seu gemmarum ossiumue sculptura auro decorat Italia, quicquid in fenestrarum pretiosa uarietate

diligit Francia, quicquid in auri, argenti, cupri et ferri lignorum lapidumque subtilitate sollers

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
1488) workshop, which specialized in painting, sculpture, casting and metal car-

pentry. One of the most important works created in this workshop when Leonardo

was still one of Verrocchio’s assistants was the copper sphere on the top of the

lantern on Filippo Brunelleschi’s (1377–1446) dome.18 In Verrocchio’s workshop,

assistants and pupils, including Leonardo, prepared colors, glues, solvents, waxes,

acids, alloys and so on. Being an artist in the Renaissance meant being part of a wider shared material culture as artisans and chymists used and developed the same materials and techniques.

In Biringuccio's book we find a description of a brass-making workshop he visited in Milan that discusses the organization of the workshop and the artisans' awareness of the technological and epistemological value of their works:

> [. . .] So that whoever entered that shop and saw the activity of so many persons would, I think, believe as I did that he had entered an Inferno, nay, on the contrary, a Paradise, where there was a mirror in which sparkled all the beauty of genius and the power of art.19

Biringuccio underlined the frenetic and very organized work in the workshop stressing the beauty of the mind of the artisans and the power of art. The harmo-nious cooperation between artisans and the plain consciousness of the processes of the art transformed the hell of hard and dirty metallurgical work into the heaven of the pursuit of the artistic goal. This oxymoron referred to the infernal and paradi-siacal condition of art, exalting the extremely difficult process in which raw material is transformed into artifact, giving back an important social value to manual work.

16 For a recent description of the chemical utensils found in the Schloss Oberstockstall laboratory, see Martinón-Torres, “Tools of Chymist”; “Probierbüchlein,” 166–9; Biringuccio,

De la
pirotechnia (1977) 47r; and Agricola, De re metallica (1556), 174–208.
17Michelangelo, Carteggio, vol. I, 20–43.
18 The sphere was three metres in diameter and was composed of several copper gores, soldered

together on the top of the dome with a solar welder. For studies on the dome of Florence and the

possible involvement of Leonardo see, Scaglia, “Studi tecnologici di Leonardo,” 6–16; Di

Pasquale, “Machinery of Construction Site”; and Galluzzi, Mechanical Marvels, 18–25, 99–116.
19 Biringuccio, Pirotechnia (1942), 72. For the Italian, see Biringuccio, De la pirotechnia (1977),

fol. 20r.

In the Renaissance technical tradition the bottega (workshop) was a place for

work and not yet a space to conduct modern scientific research. Research on technology and natural phenomena was carried out thanks to the curiosity and the free enterprise of each individual artist; workshop mentality was conservative and technical knowledge was usually applied to technologically consolidated goals. The workshop was a place to produce specific artistic, technological and pharmacological objects yet it offered artisans the tools and the experiences to venture beyond the ‘normalized activity’. We know several cases of painters involved in alchemy,

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Material and Temporal Powers at the Casino di San Marco (1574–1621) 137

similar to the one illustrated in Biringuccio’s Pirotechnia (fol. 43v), where the

techniques of shaping glass artifacts both by free blowing and by blowing into a

mold are shown. The furnace painted by Butteri suggests that he was illustrating the

glass works set up by the Venetian glassmaker Bortolo who arrived in Florence in

Fig. 4 Francesco I (left) visiting his glass work. Giovanni Maria Butteri (1570–1572) (Courtesy

Museo di Palazzo Vecchio, Florence)

1569 when the Studiolowas taking shape.31 This is confirmed by the fact that on the

right side of the painting, we see material for the construction of a new furnace and

on the left Francesco is examining finished piece of glassware. The size of the glass

works is impressive and the entrance shown in the background of the painting with

a couple of distinguished Florentine courtiers or citizens just inside the door opens

up a glimpse of the Loggiato of the Galleria della Uffizi.32 Between Stradano's

painting of the alchemical laboratory and Butteri's of the glass works, a painting by

Alessandro Fei (1543–1592) depicts a man who seems to be Prince Francesco

working on the gem-studded grand ducal crown in the atelier of a goldsmith and

jeweler (Fig. 5). The arrangement of the painting suggests that glassmaking and

jewelry were the arts most closely connected with alchemy. Their connections

inspired Francesco to create a new decorative scenario, dominated by alchemy,

by which illustrate the relation between arts and nature.

Francesco was extremely keen, even more than his father, to promote the study

of the natural sciences and the arts related to them. His passion for the

chemical arts

was so strong that in 1560, when he was only 19, he was told off by his brother

Giovanni for attending the works of the fonderia all day long.33 It was Francesco

who, in September 1569, charged the architect, engineer and inventor Bernardo

Buontalenti (1531–1608), who had been one of his teachers since 1550, to oversee

the construction of Bortolo’s new glass furnace and a new fonderia.34 Francesco I

also exploited Buontalenti’s versatile skills in the works of the fusion of precious

stones and, in particular of rock crystal. Francesco’s project to enhance his multi-

faceted interest in the alchemical and chemical arts found a new, imposing site with

the construction of the Casino di San Marco (1567–1574), a sumptuous palace

designed by Buontalenti (Fig. 6).35 Across the street from the Convent of San

31As pointed out by Heikamp. Studien, 63–9.

32 At the time the ground floor of the Uffizi was already completed. I thank Suzy Butters for

providing me with this information.

33 “Guardi di non si profonder troppo nel piacer della Fonderia, che qua vien detto, che ella non

esce mai et massimamente il giorno; talché al ritorno nostro speriamo di veder qualche nuova e

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
disciplines are embedded within the genre of the Fachliteratur.13 This kind of

literature regroups all texts of a utilitarian and informative nature whose content

does not principally concern aesthetic or religious issues, or matters relating to

emotional purpose.14 A great number of these writings share the same format and

are quite similar in terms of their external and internal characteristics. Within these compilations, the recipe frequently appears as the “shortest element in which the text could ultimately be divided.”15 This observation, although initially relating to the field of alchemy, can also be applied to recipe books in general during the Middle Ages and the premodern period. Robert Halleux underlined the similarity in format between the mediaeval treatises of alchemy and the so-called technical recipe books. He states that, whatever subject the recipe books are dedicated to, they all present a similar structure, from the earliest Mesopotamian examples to the pharmacopoeia texts of the sixteenth century. We could refine this definition

by

adding that the recipe is the smallest ‘independent’ element into which these texts

could be divided. In fact, a recipe could be seen as an independent text in itself and

could thus be dissociated from its original recipe book and be introduced into the

pages of another manuscript. For this reason, it may be argued that the recipe, as a

type text, could be considered as a structural unit common to several disciplines

embedded within the manuscripts belonging to the Fachliteratur and serve to define
a genre in itself. As Bruno Laurioux noted “[the recipe] gives the tone and

standardizes, by its repetitive structure, the corpus of this literary genre.”16

12 Smith & Hawthorne, “Mappae Clavicula.”
13 The Fachliteratur has been the subject of various studies. Concerning the German production,

see notably Eis,Mittelalterliche Fachliteratur, and more recently Haage &Wegner, Fachliteratur
der Artes.
14 Jansen–Sieben, Repertorium, XII.
15 Halleux, Textes alchimiques, 74.
16 “C’est elle qui donne le ton et uniformise, par sa structure répétitive, l’ensemble de ce genre

littéraire.” (Laurioux, Livres de cuisine médiévaux, 13).

Craft practices, alchemical treatises and artists’ recipe books thus share parts of

the same specific syntax, the frequent use of the imperative form but also some

particular verbs (such as ‘grind’, ‘mix’, ‘purify’) and vocabulary. For example, the

first alchemists used the word tinctura to refer to the tinting or the dyeing of metals,

stones or clothes.17 These methods notably explained how to dye metals yellow or

white—so (apparently) how to transmute them into gold or silver. They also

described various ways to counterfeit precious stones.18 In this context, the term

tinctura does not relate to the artisanal practice of dyeing, but instead describes the

procedure for executing the transition from one colour to another, through the steps

of the alchemical process. Another example is provided by the terms ‘mercury’ and

‘sulfur’. According to the context, these may alternatively designate the common

substances used for making vermilion or the two principles of which all metals

were

thought to be composed in different proportions.19

Thus, both artistic practices and alchemy required procedures involving precise

and specifically defined actions, prescriptions and ingredients. So both used an

identical rhetorical recipe formulation that reflects a ‘step by step’ procedure.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
they also have the same types of production and dissemination? Using a corpus of

about 40 manuscripts produced in Northern Europe between the fourteenth and the

sixteenth centuries, this paper investigates the context behind these writings, and the

various ways alchemical and artisanal recipes were embedded within recipe books.

It also proposes some clues to assist in locating, identifying and demarcating

alchemical writings within the literature of recipes.

In the Middle Ages and premodern period, alchemical knowledge and practice was

frequently transmitted via collections of recipes grouped concomitantly with artistic

instructions. Presented in the form of a succession of more or less short notes, these

S. Neven (*)

FRS-F.N.R.S, University of Liège (ULg), ‘Transitions’ Liège, Belgium

e-mail: Sylvie.Neven@ulg.ac.be

S. Dupré (ed.), Laboratories of Art, Archimedes 37,

DOI 10.1007/978-3-319-05065-2_2, © Springer International Publishing Switzerland 2014

mailto:Sylvie.Neven@ulg.ac.be

writings describe processes for the manufacture, preparation and application of

various types of materials and substances. The majority are anonymous compi-

lations of texts, which may originate from older or undetermined authorities.

Hundreds of such collections of recipes dealing both with alchemical and

art-technological procedures were produced and disseminated in Northern Europe

from the fourteenth century on, especially in German-speaking countries.

Drawing on a delimited corpus of about 40 representative German manuscripts

dated from the fourteenth to the sixteenth century, this paper investigates the

connections and similarities between these two fields and examines the various

ways alchemical and artistic instructions were embedded within recipe books.1 It

argues that textual form and lexical proximities within recipes from these different

disciplines may lead to association, contamination and confusion within this textual

genre. It finally suggests some clues to help locate, distinguish and demarcate

alchemical content within the literature of recipes.

Art and Alchemy Within Recipe Books

At first sight, any overlap between alchemy and art-technology within recipe books

can be broadly explained by the mutual use of various materials and substances

such as “common stones, gems, and types of marble, gold and other metals, sulfurs,

salts, and inks, azures, minium, and other colors, oils and burning pitches,

and

countless other things.”2 More precisely, the field of art-technology encompasses a

large range of craft practices involved in the production of pieces of art (including

those which incorporate such substances). This ‘hand’ knowledge, is related to the

mechanical arts and is divorced from the philosophical or speculative dimension.

Yet, alchemy could be described as the practical, philosophical and medical search

for the perfecting of base material substances and also for the extending of life.3

The theoretical and practical aspects of alchemy involve both the study of all

inanimate or animate things made from the elements and the observation and

imitation of natural processes within the laboratory.4 In this context,

alchemy

could be seen as a mechanical art, in the sense that it works on matter but is also

a liberal art, as it attempts to explain matter in its composition and its transforma-

tion.5 In the practical sense, one of the main goals of alchemy is the transmutation

1 The main data and characteristics of these manuscripts are given in Appendix.
2 Brewer, Fr. Rogeri Bacon, 39–40.
3 Perception and definition of alchemy is not chronologically constant and has been the subject of

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
different recipes.

As stated above, whether alchemical or art-technological, the recipes contained

in these manuscripts are presented in the form of a formula which, in most cases,

enumerates the ingredients and the actions necessary to produce a particular

preparation. In addition artistic recipes sometimes indicate the recommended

geographical provenance or grade of quality of the ingredients. Suggestions for possible substitutions might also appear. This sort of information is rare in alchemical recipes.

The length of a recipe depends not only on the number of ingredients involved but also on its complexity, the number of steps necessary to obtain the final product.

A recipe can be anything from one sentence to several pages within a manuscript.

Alternatively, a recipe may appear merely as a brief list of ingredients, without any other additional information. In fact, two categories of recipe can be distinguished: the Vollrezepte (detailed recipes) and the Kurzrezepte (abbreviated recipes).55 In

the first, the quantities and the various steps are indicated. In the second, only the

55Halleux, “Alchimie,” 343.

ingredients are cited: the procedure is sketched out or omitted altogether and the

rest is left to the ingenuity of the user. This second category is more common in the

case of artistic recipes; a great many of the recipes dedicated to the manufacture of

ink are written in the form of a very short list of ingredients. It is less common for

the alchemical recipes to be presented this way.

The title of a recipe may also give an indication of the final product to be obtained

and, in some cases, specify the use of the product. Again, this is particularly true for

artistic instructions and is less observable for alchemical ones.

For both types of instructions (alchemical or artistic), some steps could be

omitted or were left to the interpretation of the reader. Specified quantities may

be missing in both fields. When quantities are given, artistic recipes are far more

likely to use local measurements, whereas in alchemical instructions, the quanti-

ties—if mentioned at all—are more often expressed in terms of ratio or proportions.

In some cases, these proportions are not ‘practically’ correct. A very well-known

example is the proportion of mercury and sulphur proposed by mediaeval recipes

for the production of vermilion which is invariably incorrect.56 Very rarely are the correct chemical proportions cited.

In parallel, alchemical writings may involve the use of symbols or metaphors to designate substances and practices. In consequence, the way an alchemical recipe was received would depend on the degree of experience of the reader-practitioner reading it. On the one hand, the (sometimes) metaphorical or codified language as well as the approximations stressed the arcane nature of these recipes and contributed to their secrecy. On the other hand, the omitted information may have been complemented by data only known to some readers and not recorded by the copyist who conserves only the essential part of the recipe. If so then this kind

of

recipe was only meant to be accessible and useable by those practitioners who could

easily fill in the lacuna that punctuated the text of the recipe.

As previously observed, citations of authority were frequently used by the scribes

of the manuscripts. However, the tendency for an older authority to be cited in the

recipe books is particularly characteristic of the alchemical writings and less typical

of the art-technological recipes. As stated above, such citations primarily served to

legitimate the technical and chemical procedures. In addition, most alchemical

recipes describe processes and practical results to validate previously enounced

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
l’avesse tanto avvilita, con ragioni assai deboli, come quegli, il quale avea molta pratica, e non

molta scienza, poiché egli non sapeva, lasciamo stare degli altri, ma che né Alberto ancora

n’avesse scritto; ma comunque sia, gli semo obbligati grandissimamente essendo stato, oltre la

grandissima pratica, uomo molto leale e veritiero, e liberalissimo dei suoi tesori.” (Varchi,

Questione sull’alchimia, 63–4).
10 Bernardoni, Conoscenza del fare, 31–3.

alchemy and from uncontrolled techniques. Both of these authors gave credit to

alchemy for producing codified procedures to transform substances and, although it

is extremely difficult to talk about their role in the development of chemical

technology, we do not have any doubts about their important contribution as pro-

tagonists and witnesses of the epistemological and technological transformations in

the field of ‘science and technology of matter’.

If we focus our attention on all the artisanal processes of substance transforma-

tion known and practiced during the fifteenth and sixteenth centuries it is easy to

show that they do not fit perfectly within the domain traced by alchemy. This

epistemological tension is usually neutralized by saying that alchemical and met-

allurgical disciplines are intricately interwoven and alchemists were involved in the

development of metallurgy.11 If one supposes chemistry did not exist as an inde-

pendent and autonomous discipline before the end of the seventeenth century then

speaking on this topic from a historical perspective in an Italian technical cultural

context, some ‘superior artisans’ or ‘artist-engineers’ can be seen as having dis-

tanced themselves from alchemy by relocating its technology inside a new

episte-

mological context in which chemical arts were developed and codified separately.12

This context applies not only to metallurgical assaying but, more generally, it

defines the area of the ‘perfective arts’, including all arts that use fire to work and

transformmatter.13 In this trading zone, where Latin was scarcely known and where competition with alchemists was felt more keenly, a sub-group of artisans, artists

and engineers kept their distance from alchemy to define a new field of ‘chemical

technology’.14

Artisanal processes could be seen as a sort of ‘melting pot’ of interwoven

practical and theoretical knowledge coming from the world of crafts, including

alchemy. It is prevalently in the world of artisans that we find references to ‘chemical activity’ during the sixteenth and seventeenth centuries. Before there were signs of a laboratory as a separate place—the most ancient plans to build a chemical laboratory are in the Astronomiae instauratae mechanica (Wandsbek, 1598) by Tycho Brahe (1546–1601) and in the Commentariorum alchymiae (Frankfurt, 1606) by Andreas Libavius (1560–1616), ‘chemical operations’ could be found everywhere ‘perfective arts’ were practiced.15 This was in alchemist’s laboratories, such as the one in Schloss Oberstockstall (Austria) in the sixteenth century, and assaying workshops, described in the anonymous Probierbuchlein

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Strohmaier, Gotthard. 1991. Umāra ibn H
˙
amza, Constantine V and the Invention of the Elixir.

Graeco-Arabica 4: 21–24.

Westerink, Leendert G. 1948. Michael Psellus, De Omnifaria Doctrina. Nijmegen: Centrale

Drukkerij N.V.

22 M. Martelli

Transmission of Alchemical and Artistic

Knowledge in German Mediaeval

and Premodern Recipe Books

Sylvie Neven

Abstract In the Middle Ages and premodern period, artisanal knowledge was transmitted via collections of recipes often grouped concomitantly with alchemical texts and instructions. Except for some very well-known artistic treatises, e.g. works by Eraclius or the Schedula diversarum artium, attributed to Theophilus, detection and delimitation of alchemical content within recipe books has been rare and fraught with difficulty. Alchemy can be defined as the ‘art of transmutation’, referring to the perfection of base or impure matter (often metal or stone) into perfect substances. Alchemical procedures thus rely on artisanal/craft practices.

Any overlap between alchemy and art-technological procedures can be explained by the use of identical materials and substances. Both are concerned with the description of colours—especially in processes of change, the making of pigments, the production of artificial gemstones, the imitation of gold and silver and the transmutation of materials. Both require procedures involving precise and specifi-cally defined actions, prescriptions and ingredients. So both ultimately use identical rhetorical formulations that reflect a ‘step by step’ procedure. Assuming that alchemical and artistic texts have the same textual format, raises the question: did they also have the same types of production and dissemination? Using a corpus of

about 40 manuscripts produced in Northern Europe between the fourteenth and the

sixteenth centuries, this paper investigates the context behind these writings, and the

various ways alchemical and artisanal recipes were embedded within recipe books.

It also proposes some clues to assist in locating, identifying and demarcating

alchemical writings within the literature of recipes.

In the Middle Ages and premodern period, alchemical knowledge and practice was

frequently transmitted via collections of recipes grouped concomitantly with artistic

instructions. Presented in the form of a succession of more or less short notes, these

S. Neven (*)

FRS-F.N.R.S, University of Liège (ULg), ‘Transitions’ Liège, Belgium

e-mail: Sylvie.Neven@ulg.ac.be

S. Dupré (ed.), Laboratories of Art, Archimedes 37,

DOI 10.1007/978-3-319-05065-2_2, © Springer International Publishing Switzerland 2014

mailto:Sylvie.Neven@ulg.ac.be

writings describe processes for the manufacture, preparation and application of

various types of materials and substances. The majority are anonymous compi-

lations of texts, which may originate from older or undetermined authorities.

Hundreds of such collections of recipes dealing both with alchemical and

art-technological procedures were produced and disseminated in Northern Europe

from the fourteenth century on, especially in German-speaking countries.

Drawing on a delimited corpus of about 40 representative German manuscripts dated from the fourteenth to the sixteenth century, this paper investigates the connections and similarities between these two fields and examines the various ways alchemical and artistic instructions were embedded within recipe books.1 It

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
11 Halleux, “Alchimiste et l’essayeur.”; and Nummedal, Alchemy and Authority, 85–91.
12 For more on the technical cultural context in the history of science and technology, see Zilsel,

Social Origins of Modern Science; Maccagni, “Leggere, scrivere e disegnare”; Galluzzi, “Portraits

of Machines; Smith, Body of the Artisan; Halleux, Savoir de la main, 102–39; and Long, Artisan/
Practitioners, 10–29.

13 Newman, Promethean Ambitions, 17–20.
14 Long, Artisan/Practitioners, 94–6. See also, Galison, “Trading Zone.”
15 Hannaway, “Laboratory Design.”

(1513), in Biringuccio’s Pirotechnia, and in Agricola’s De re metallica.16 Further nascent signs of a distinct and autonomous chemical laboratory with purpose-built ‘chemical’ equipment can be found in the more general context of the Renaissance artist’s workshop culture. Here, together with the fine arts such as painting, sculpture, architecture and jewelry making, metal carpentry, and gun casting, we find operations that use substance transformation, such as distillation and sublimation.

There is evidence that even the most important names in the history of art were

involved in such workshop activities, for instance Michelangelo (1475–1564), whose Florentine workshop, as mentioned in his correspondence with his brother, was involved in generic metal carpentry such as the casting and welding of damaged swords and other craft objects.17 Further evidence of artists' versatile activities during the Renaissance can be found in Andrea Verrocchio's (1435–1488) workshop, which specialized in painting, sculpture, casting and metal car-pentry. One of the most important works created in this workshop when Leonardo was still one of Verrocchio's assistants was the copper sphere on the top of the lantern on Filippo Brunelleschi's (1377–1446) dome.18 In Verrocchio's workshop, assistants and pupils, including Leonardo, prepared colors, glues, solvents, waxes,

acids, alloys and so on. Being an artist in the Renaissance meant being part of a wider shared material culture as artisans and chymists used and developed the same materials and techniques.

In Biringuccio's book we find a description of a brass-making workshop he visited in Milan that discusses the organization of the workshop and the artisans' awareness of the technological and epistemological value of their works:

> [. . .] So that whoever entered that shop and saw the activity of so many persons would, I think, believe as I did that he had entered an Inferno, nay, on the contrary, a Paradise, where there was a mirror in which sparkled all the beauty of genius and the power of art.19

Biringuccio underlined the frenetic and very organized work in the workshop stressing the beauty of the mind of the artisans and the power of art. The harmo-nious cooperation between artisans and the plain consciousness of the processes of the art transformed the hell of hard and dirty metallurgical work into the heaven of the pursuit of the artistic goal. This oxymoron referred to the infernal and paradi-siacal condition of art, exalting the extremely difficult process in which raw material is transformed into artifact, giving back an important social value to manual work.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
several (re)interpretations since the eighteenth century, see Principe & Newman, “Historiography

of Alchemy.”
4Halleux, “Alchimie,” 336–7; Newman, “Technology and Alchemical Debate,” 432–3; Pereira,

“Use of Vernacular Languages,” 336; and Kahn, Alchimie et paracelsisme, 7–8.
5 Halleux, Savoir de la main, 134.

24 S. Neven

of base or impure matter (often metal or stone) into a noble or perfect one.6 To do

so, alchemists used to perform chemical processes and manipulations which resem-

bled those practiced by contemporary artists and artisans.

Thus, in both fields particular importance is placed on craft practices. Both

alchemical and artistic recipe books describe various processes for purifying and

transforming materials, either for improving their properties or in order to use them

for specific purposes. In this context, from a technological point of view, the term

‘transmutation’ could refer to the colouring of glass, the melting and tinting of

metals, the dyeing of stones or gems, or the manufacture of synthetic pigments. A

huge number of recipes are dedicated to procedures for obtaining gold or silver or

gold-silver like substances from base metals (copper, tin, lead, iron, mercury).

The tradition of recipe books has roots deep in Antiquity. Treatises dealing both

with art-technological and alchemical procedures notably survived in the Alexan-

drian Greek papyri preserved in Leyden and Stockholm.7 These date from the third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

They also have recipes dedicated to the art of dyeing, to chrysography, and to the

making of artificial precious stones.

An echo of these recipes can be found in the Codex Lucensis 490 (Lucca,

Biblioteca Capitolare Felinià na), also known as the Compositiones ad tingenda musiva or Compositiones variae. This manuscript, dated from the end of the eighth

or the very beginning of the ninth century, is based on previous Greek sources

compiled around the beginning of the seventh century.9 This recipe book deals with

various artistic techniques, notably the dyeing of skins, the manufacture of pig-

ments, colorants, varnishes and glues, chrysography and the gilding of metals.

The Lucca Manuscript shares content with the Mappae clavicula, compiled

around 800. Parts of this text have far earlier origins and present parallels not

only with the Leyden and Stockholm papyri but also with Syriac translation of

Zosimus and with ps.–Democritus’s writings.10 The nucleus of this text was probably a Greek alchemical treatise written and translated between the fourth and the fifth centuries, with additions in the eleventh and twelfth centuries.11 The two main manuscripts are the Corning, Corning Glass Museum, Ms. 5 and Sélestat, Bibliothèque Humaniste, Ms. 17, but this tradition was also (partly) disseminated through dozens of manuscripts. The Mappae clavicula contains about 300 recipes

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
equipment from the Casino to the Uffizi. During Francesco I’s reign the Casino housed a very important artistic glass workshop where the prince worked on his

own experiments (for example smelting rock crystal).29 This laboratory was lead by

Sisti, who also lead a glassmaking laboratory in Pisa, where he often had to go. In

his early career he used the technique a lume di lucerna, but once he transfered

to

the Uffizi, this activity became secondary.30

At the Uffizi, glass production may well have been located in the smithy

workshop. It was probably limited to the glass technique a lume di lucerna. The oven glass manufacturing method came from existing production centres such as

the ones in Pratolino, or Pisa, which became the most important in Tuscany. On the

other hand, during Ferdinando I’s reign, documents already show activities by

someone called Niccolò di Vincenzo Landi di Lucca, after Sisi, the principal

glassmaker in service to the Medici’s till 1620. The sources show that Landi, a

specialist in a lume di lucerne, went on duty in the fonderia of the Uffizi in 1591,

where he made little animals for the decoration of glass manufactured in

Pisa.31 In

1601, Antonio Neri also mentioned him as the leader of the new glasswork in the

Casino di San Marco.32 Later, in 1618, when Grand Duke Cosimo II (1590–1621)

set up a large workshop with many ovens in the Boboli gardens near the Pitti palace,

28 ASF, GM 403, ins. 2, fol. 120.
29 On the story of the Medicean glassmaking, see Heikamp, “Mediceische Glaskunst.”
30 The technique a lume di lucerna allows to make or decorate little objects heating locally the

glass elements thanks to a lantern flame or a candle.
31 ASF, GM 112, passim; GM 217, fol. 23; and GM 195, ins. 1, fol. 102.
32 On the priest Antonio Neri and his treatise on the art of glassmaking, see Abbri’s introduction to

Neri, L’arte vetraria (2001), 5–23.

he entrusted its management to him, which shows that he was certainly able to

supervise large-scale production.

The fonderia produced mainly small ornamental glass items (Fig. 7). The items

in use for distillery and for the other workshops came either from Pisa, or from the

Casino di San Marco.33 Mainly made of bronze or metal, the models of these little

items were created by the court goldsmiths or by the smelters: there were little

masks, buttons to be filled with perfumes, ornamental glasses.34 The glasses, stored

in a hall of the Galleria “la stanza dei cristalli,” were distributed according to the

grand duke’s wishes and other workshops requirements (most of them were given

as presents or used to decorate other items coming from other workshops).35 Sisti,

in charge of the fonderia, was responsible for the seamless transitions between the

different stages of the process, which was not always an easy task in the case of a

delicate material such as crystal.36

The Forge

In the Uffizi, the forge and smelting works were far less visible and have less

prestige. Only small items are worked on in the forge: little masks for the orna-

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
Böttger Porcelain 1708 61.0 33.0 0.00 4.8 0.00 0.1 0.2 0.9 100

Meissen Porcelain (ft48)

mid-eighteenth century

59.0 35.00 0.00 0.3 0.00 4.0 0.00 0.9 100

198 M. Wesley

Bibliography

Bacon, Francis. 1858. New Organon, trans. James Spedding et al. London: Longmans and Co.

Birch, Thomas. 1756–1757. History of the Royal Society of London. London: A. Millar.

Biringuccio, Vannoccio. [1540] 1942. The Pirotechnia of Vannoccio Biriniguccio: The Classic Sixteenth-Century Treatise on Metals and Metallurgy, eds. and trans. Cyril Stanley Smith and

Marta Teach Gnudi. New York: Dover Publications.

Boardman, John. 1988. Trade in Greek Decorated Pottery. Oxford Journal of Archaeology 7: 27–33.

Boyle, Robert. 1661. The Sceptical Chymist or Chymico-Physical Doubts and Paradoxes, Touching the Spagyrists Principles Commonly call'd Hypostatical: As They Are Wont To Be Propos'd and Defended by the Generality of Alchymists. Whereunto is Premis'd Part of Another Discourse Relating To the Same Subject. London: J. Cadwell for Crooke.

Boyle, Robert. 1691. Will of the Hon. Robert Boyle, F.R.S, July 1691. The National Archives,

Manuscript PROB 11/408/169.

Carswell, John. 2000. Blue and White: Chinese Porcelain Around the World. London: British
Museum Press.

Chaffers, William. 1870. Marks and Monograms on Pottery and Porcelain, 3rd ed. London: J

Davy and Sons.

Charles the Second. 1672. Patent for the Protected Manufacture of Transparent Earthenware.
Public Record Office. C. 82 2425 Cal. S. P. Dom Entry Book 34.

Cruz, Gaspar da. 1569. Tractado em que se cotam muito por esteso as cousas da China. Madrid:

em casa de Andre de Burgos.

Debus, Allen G. 1967. Fire Analysis and the Elements in the 16th and 17th Centuries. Annals of
Science 23: 127–147.

d’Entrecolles, Père Francois Xavier. 1781. Lettre D’Entrecolles à Jao-tcheou, 1er Septembre

1712’ and ‘Lettre D’Entrecolles à Kim-te-tchim, le 25 Janvier 1722. In Lettres édifiantes et
curieuses écrites des missions
estranges. Mémoires de la Chine etc., vol. 18–19. Paris: Societas
Jesu Missio Sinensis.

Foster, Joseph. 1891. Alumni Oxonienses 1500–1714. London: Parker and Co.

Green, Chris. 1999. John Dwight’s Fulham Pottery: Excavations 1971–1979. London: English
Heritage.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
argues that textual form and lexical proximities within recipes from these different

disciplines may lead to association, contamination and confusion within this textual

genre. It finally suggests some clues to help locate, distinguish and demarcate

alchemical content within the literature of recipes.

Art and Alchemy Within Recipe Books

At first sight, any overlap between alchemy and art-technology within recipe books

can be broadly explained by the mutual use of various materials and substances

such as “common stones, gems, and types of marble, gold and other metals, sulfurs,

salts, and inks, azures, minium, and other colors, oils and burning pitches, and

countless other things.”2 More precisely, the field of art-technology encompasses a

large range of craft practices involved in the production of pieces of art (including

those which incorporate such substances). This ‘hand’ knowledge, is related to the

mechanical arts and is divorced from the philosophical or speculative dimension.

Yet, alchemy could be described as the practical, philosophical and medical search

for the perfecting of base material substances and also for the extending of life.3

The theoretical and practical aspects of alchemy involve both the study of all inanimate or animate things made from the elements and the observation and imitation of natural processes within the laboratory.4 In this context, alchemy could be seen as a mechanical art, in the sense that it works on matter but is also a liberal art, as it attempts to explain matter in its composition and its transformation.5 In the practical sense, one of the main goals of alchemy is the transmutation

1 The main data and characteristics of these manuscripts are given in Appendix.
2 Brewer, Fr. Rogeri Bacon, 39–40.
3 Perception and definition of alchemy is not chronologically constant and has been the subject of

several (re)interpretations since the eighteenth century, see Principe & Newman, “Historiography of Alchemy.”

4Halleux, “Alchimie,” 336–7; Newman, “Technology and Alchemical Debate,” 432–3; Pereira, “Use of Vernacular Languages,” 336; and Kahn, Alchimie et paracelsisme, 7–8.

5 Halleux, Savoir de la main, 134.

of base or impure matter (often metal or stone) into a noble or perfect one.6 To do so, alchemists used to perform chemical processes and manipulations which resembled those practiced by contemporary artists and artisans.

Thus, in both fields particular importance is placed on craft practices. Both alchemical and artistic recipe books describe various processes for purifying and

transforming materials, either for improving their properties or in order to use them

for specific purposes. In this context, from a technological point of view, the term

‘transmutation’ could refer to the colouring of glass, the melting and tinting of

metals, the dyeing of stones or gems, or the manufacture of synthetic pigments. A

huge number of recipes are dedicated to procedures for obtaining gold or silver or

gold-silver like substances from base metals (copper, tin, lead, iron, mercury).

The tradition of recipe books has roots deep in Antiquity. Treatises dealing both

with art-technological and alchemical procedures notably survived in the Alexan-

drian Greek papyri preserved in Leyden and Stockholm.7 These date from the

third

century but were probably based on previous texts.8 These papyri contain informa-

tion on the imitation of precious materials such as gold, gems and Tyrian purple.

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
porary artists and practicing scholars or from the scribe’s own experiments, as the

examples of Seidel, Freising and Schobinger discussed above illustrate. Even when

the writing of these instructions, verbalized in the rhetoric of the recipes, was

carried out by scribes, data were not blindly copied. Scribes organised, assembled,

completed or corrected when they felt it necessary. Thus, even if they were not the

author per se, in the sense that they were not the origin or the source of the technical
or chemical procedures they wrote down, they accomplished a set of activities

linked to ‘authorship’.50 Scribes also made attempts to ensure that the recipes could

be consulted at need: they composed tables of contents or indexes, they introduced

titles within the margins and many other details which attest to a real desire to

deliver usable information. In this context the marginal notes and additions made by

the scribes/authors of the recipe book are of interest as most of them are technical

comments testifying practical interest in both alchemical and artistic instructions.

Several marginal annotations due to Seidel’s hand punctuate the Cgm 4117 and

consist in personal commentaries regarding the technical procedures he records.

For example, on folio 53r, Seidel compares two ways for the melting of crystal.

Concerning the first process he states in the margin that this ‘art’ was not of use to

him as a better (method) is delivered on 219.51 Then on folio 219, he indicates

another method for the same technical procedure, giving as title “How one should

masterfully melt crystal.”52

In this sense, the scribes at the root of these recipe books created not simply a

copy but a unique work, which reflected their own interests, their cultural and life

context and sometimes their intention, which was to deliver practical and useful

instruction.

49 Halleux, Entre technologie et alchimie, 7.
50 For this definition of authorship, see notably Love, Attributing Authorship, 32–40.
51 “Dise kunst prauchet ich nit hinden amm 219 hastu vil pessere.” (Munich, Bayerische Staatsbi-

bliothek, Cgm 4117, fol. 53r)
52 “Wie man christallen maisterlich giessen soll.” (Munich, Bayerische Staatsbibliothek, Cgm

4117, fol. 219)

Reliability of Recipe Books

The modalities of composition and diffusion of these recipe books have an impact

on their current (practical) use. During the compiling and disseminating processes,

both alchemical and art-technological collections of recipes were subject to muta-

tions, in the form of interpolation, reduction, contamination or assimilation with

other texts. As the recipe books evolved and were modified by adding new texts and

procedures, the recipes themselves could be modified in their technical formula-

tions during their transmission from one manuscript to another. Assimilation with

other texts occurs quite frequently, as the ingredients (and the actions) specified in

these texts appear in the artistic recipe books but also in medical treatises, cookery

books, and in alchemical or magical texts. Frequently, the copyist was free to add,

to remove or to omit some words or even some parts of the text. These modifica-

tions or omissions sometimes concern primary data, such as the name of the ingredients or materials, or may be related to some of the steps of the procedure.

At each stage of the copying process, variations or errors can occur. This pheno-

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
application of its techniques to reverse the process of making porcelain is a logical

extension of the economic demand for china and the intellectual drive toward

maker’s knowledge as expressed by Bacon.

Certainly, the history of ceramics after the seventeenth century is inextricably

linked to chemistry’s success at solving increasingly sophisticated questions of

luxury pottery production. We need look no further than William Cookworthy’s

(1705–1780) patent for the manufacture of porcelain in 1768 or the thousands of experiments conducted by Josiah Wedgwood (1730–1795) in the production of his Jasperwares. Unlike the other arts of the fire; metalworking, glassmaking, and dyeing, the late inclusion of ceramics into the hierarchy of decorative arts allows contemporary scholars opportunities to investigate the separations between the craft artisan and the chymist during the early modern period, of which this discus-sion of thermal analysis is only one of many.

Table 1 A comparison of modern elemental analyses of ‘Dwight’ experimental ware, Blanc-de-Chine, ‘Böttger’ ware, and Kangxi porcelain

Material	SiO_2	Al_2O_3	Fe_2O_3	CaO	MgO	K_2O	Na_2O	Other	Total

Dwight 80.3 12.4 0.74 0.39 0.2 5.35 0.47 Trace 99.1

Blanc-de-Chine (ft46) 76.7 16.8 0.35 0.15 0.08 5.9 0.2 Trace 100

Kangxi Porcelain (ft47) seven-

teenth century

64.7 28.35 0.95 0.5 0.1 2.8 2.4 0.19 99.99

Böttger Porcelain 1708 61.0 33.0 0.00 4.8 0.00 0.1 0.2 0.9 100

Meissen Porcelain (ft48)

mid-eighteenth century

59.0 35.00 0.00 0.3 0.00 4.0 0.00 0.9 100

Bibliography

Bacon, Francis. 1858. New Organon, trans. James Spedding et al. London: Longmans and Co.

Birch, Thomas. 1756–1757. History of the Royal Society of London. London: A. Millar.

Biringuccio, Vannoccio. [1540] 1942. The Pirotechnia of Vannoccio Biriniguccio: The Classic Sixteenth-Century Treatise on Metals and Metallurgy, eds. and trans. Cyril Stanley Smith and Marta Teach Gnudi. New York: Dover Publications.

Boardman, John. 1988. Trade in Greek Decorated Pottery. Oxford Journal of Archaeology 7: 27–33.

Boyle, Robert. 1661. The Sceptical Chymist or Chymico-Physical Doubts and Paradoxes, Touching the Spagyrists Principles Commonly call'd Hypostatical: As They Are Wont To Be Propos'd and Defended by the Generality of Alchymists. Whereunto is Premis'd Part of Another Discourse Relating To the Same Subject. London: J. Cadwell for Crooke.

Boyle, Robert. 1691. Will of the Hon. Robert Boyle, F.R.S, July 1691. The National Archives,

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:
individuals and evidence presented in the archaeological record demonstrates

Dwight’s re-firing of Chinese sherds while Tschirnhaus published results of the

experimental application of burning lenses to determine the physical properties of

varied ceramic bodies. Dwight was ultimately unsuccessful in his endeavors, likely

due to economic realities, while Tschirnhaus’s work would lay the cornerstones for

the eventual production of Meissen porcelain.

The Potter’s Art as Craft Practice

To understand why the shift from the intellectual space of the craft workshop to the

chymical laboratory was critical to the advancement of European ceramic devel-

opment, an examination of the awkward position of pottery within the hierarchy of

the decorative arts before the early modern period is worthwhile. Prior to its

interaction with chymistry, the relationship between the potter’s craft and fire was

principally of a mechanical, fixative nature. As the heat of the kiln was known to

remove the water from earthen bodies, its ability to affect permanent transformation

on the materials placed in the kiln was its principal contribution to the success of

ceramic production. Those objects existed first as trade goods rather than artisanal

objects and only later became valued as aesthetic products. Vannoccio

Biringuccio's (1480–c.1539) introduction to his brief "discourse on the art of the

potter" from his Pirotechnia of 1540, encapsulates both the recent recognition

received by ceramic craft and its inseparability from fire:

Igne refutata: Thermal Analysis in the Laboratory Practices of John Dwight. . . 183

Having started to tell you of working potter's clay for making crucibles and shells, the wish

came to me to tell you of the practice of this art also. Although it may seem at first glance to

be outside the order and purpose of my writing, he who considers well will see that it is not

unrelated to it but proper, since it is wholly dependent on the agency and power of fire if it is

to be brought to its perfection. Moreover, the potter’s glazes and colors are all substances of

various metals or impure minerals and therefore belong to fire. Since it is my intention to

treat of fire, minerals, and metals for you, I surely should not have omitted this, particularly

because it is a necessary art which enriches and is greatly praised both for its ingenuity and

its beauty.4

Like the other arts treated in the Pirotechnia, the action of the fire on base earths

and minerals provided integral transformation in the objects shaped by the potter's

hands. At the time of Biringuccio's writing, the potter's art was on the threshold of a

paradigm shift that would create a marked division between craft knowledge and

theory based practice, setting two separate courses for ceramic innovation in the

later sixteenth and seventeenth centuries. His citation of alchemy as one of "two

sources as [pottery's] principal basis," is referential to the provision of purified

minerals and "elemental mixtures."5 However, it also foreshadowed the intellectual

engagement of chymists in the quest for the production of European porcelain,

bringing the weight of the older alchemical tradition to bear.

Prior to the fourteenth century, European pottery was external to the hierarchy of

the esteemed arts, such as metallurgy, glassmaking, painting, and dyeing. The

extensively copied treatise De diversis artibus, composed in the thirteenth century

by the Benedictine monk Theophilus, thoroughly introduces the full range of

artisanal productions valued in Europe in its first book.6 Pottery is entirely absent

Content source: Source.file
Title: Laboratories of Art_ Alchemy and Art Technology from Antiquity to the 18th Century-Springer (20.pdf
Content Snippet:

Paolozzi Strozzi, Beatrice, and Dimitrios Zikos. 2006. Giambologna, gli dei, gli eroi. Florence: Giunti.

Pelli Bencivenni, Giuseppe. 1779. Saggio istorico della Real Galleria di Firenze, vol. 2. Florence: Per Gaet. Cambiagi stamp. Granducale.

Perifano, Alfredo. 1997. L’Alchimie à la Cour de Côme Ier de Médicis: savoirs, culture et politique. Paris: Honoré Champion.

Pieraccini, Gaetano. 1924–1925. La Stirpe de'Medici di Cafaggiolo: saggio di ricerche sulla trasmissione ereditaria dei caratteri biologici, vol. 2. Florence: Vallecchi.

Pozzi, Mario. 2004. Filippo Pigafetta consigliere del principe. Vicenza: Biblioteca Civica

Bertoliana.

Schepelern, H.D. 1985. Natural Philosophers and Princely Collectors: Worm, Paludanus and the

Gottorp and Copenhagen Collections. In The Origins of Museums: The Cabinets of Curiosities in Sixteenth and Seventeenth-Century Europe, eds. Oliver Impey and Arthur MacGregor, 121–

127. Oxford: Clarendon.

Somers Cocks, Anna. 1980. Princely Magnificence: Court Jewels of the Renaissance, 1500–1630. London: Debrett's Peerage.

Targioni Tozzetti, Giovanni. 1769–1779. Selve di notizie, spettanti all'origine

de’ progressi e
miglioramenti delle scienze fisiche in
Toscani, messe insieme dal Dottor
Giovanni Targioni-
Tozzetti, per uso del dottor Ottaviano
suo figlio, 17 vols. Biblioteca
Nazionale Centrale di

Firenze, Targioni. Tozzetti, Ms. 189.

Tongiorgi Tomasi, Lucia, and Alessandro
Tosi. 1990. “Flora e Pomona”:
L’orticoltura nei
disegni e nelle incisioni dei secoli
XVI–XIX. Florence: Olschki.

Wazbinsky, Zygmunt. 1994. Il Cardinale
Francesco Maria Del Monte, 1549–1626, 2
vols.

Florence: Olschki.

Zobi, Antonio. 1841. Notizie storiche
riguardanti l’Imperiale e Reale
Stabilimento dei lavori di
commesso in pietre dure di Firenze.
Florence: Tip. di Felice Le Monnier.

Zorzi, Elvira Garbero, and Mario
Sperenzi. 2001. Teatro e spettacolo

nella Firenze dei Medic.
Florence: Olschki.

The Laboratories of Art and Alchemy at the Uffizi Gallery in Renaissance. . . 127

Material and Temporal Powers at the Casino

di San Marco (1574–1621)

Marco Beretta

Abstract Built in 1574 by court engineer and architect Bernardo Buontalenti for

Francesco I de Medici, the Casino di San Marco represents a unique example of a

late Renaissance site of alchemical research, art collecting and political court.

Francesco I’s program to enhance the chemical arts and make it into a body of

```
Content source: Source.file
Title: Laboratories of Art_ Alchemy and
Art Technology from Antiqu
Visible: 0% - 95%
```

ASSISTANT

Here’s a list of
mentioned in *Laboratories of Art: Alchemy and Art Technology from Antiquity to the 18th Century*. These reflect overlaps between alchemy and artisanal practice in workshops, recipe books, and princely laboratories:

- – Applying gold to surfaces 【16:11†source】.
- – Writing with gold 【16:11†source】.
- – Coloring metals to imitate silver 【16:11†source】.
- – Shaping metals in molds 【16:10†source】.
- – Metal carpentry, sword repair, sculpture supports 【16:10†source】.
- – Testing purity of metals, overlap with alchemy 【16:8†source】.
- – Creating new metal mixtures (bronze, brass, pewter).

- – Decorating metals【16:11†source】.

- – Shaping glass vessels【16:4†source】.
- – Clear, rock-crystal-like Venetian glass【16:12†source】.
- – Shaping glass with a flame【16:12†source】.
- – Adding metallic oxides for hues【16:14†source】.

- – Using colored glass to mimic rubies, emeralds, sapphires 【16:11†source】.
- – Decorative and alchemical experiments 【16:4†source】.

- – Biringuccio connects pottery to alchemy 【16:17†source】.
- – Luxury Italian ceramics 【16:2†source】.
- – Medici porcelain, Böttger ware, Meissen

【16:16†source】.

- – Using metallic oxides and fire 【16:17†source】.

- – Applying colored vitreous layers 【16:2†source】.

- – Preparing azurite, minium, vermilion, etc.【16:5†source】.

- – Oils, gums, waxes【16:3†source】.

- – Vermilion (mercury + sulfur), lead-tin yellow

【16:11†source】.

- – Protective and coloristic layers 【16:11†source】.

- – Recipes for black, colored, and gold inks 【16:7†source】.

- – With indigo, madder, saffron, Tyrian purple【16:1†source】.

- – Alchemical and artisanal overlap 【16:5†source】.

- – Fixing dyes with alum, salts.
- – Mentioned in medieval recipe books 【16:11†source】.

- – Artistic and alchemical interests 【16:4†source】.
- – Florentine art of commesso 【16:18†source】.
- – Shared metallurgical knowledge 【16:10†source】.

- – Lime-based decorative processes.

- – Using rosewater, amber, musk, herbs【16:10†source】.
- – Powders, skin dyes, ointments.
- – Court spectacles of edible art【16:9†source】.
- – Preserving fruits and medicines【16:10†source】.

- – Court displays, overlap with alchemy 【16:8†source】.
- – Functional and artistic metallurgy 【16:10†source】.
- – Shared processes of heating, sublimation, calcination 【16:2†source】.

📌 These processes show how , all depending on material transformation through . Recipe books bundled together instructions for

, blurring boundaries between .

Would you like me to (Process | Materials used | Alchemy connection | Artistic use)? That could mirror the material tables I made earlier.

USER

give me some ideas for a game simulating the involvement of early modern alchemists in artisanal culture and materials science, including social activities like attaining patronage, work and commercial activity, counterfeiting and all the various trades that alchemical knowledge is relevant to. give me 40 elevator pitches in different genres

ASSISTANT

Here are , each imagining a different way to simulate the

involvement of in artisanal culture, material science, and the social worlds of patronage, trade, and counterfeiting. I've split them across genres so you get a variety of approaches:

---

1. – Players bid their alchemical recipes to win noble patronage; the trick is proving your worth without being exposed as a fraud.
2. – A worker-placement game where you allocate apprentices to dyeing, glassmaking, or metallurgy, racing to complete commissions.
3. – Deduction-based board game where players forge gems or coins, trying not to be unmasked by rival alchemists.
4. – Card drafting game where you assemble recipes from shared pools, then sell to guilds, courts, or apothecaries.
5. – A push-your-luck game where you mix saltpeter, charcoal, and sulfur to impress patrons, risking explosions.

6. – An area-control board game where guilds of goldsmiths, glassblowers, and alchemists vie for city influence.
7. – A cooperative card game where players reconstruct fragmented manuscripts into functioning artisanal recipes.
8. – Semi-coop game where one player is secretly sabotaging Medici porcelain experiments.
9. – An economic board game about balancing real products (dyes, medicines) against speculative ventures (transmutation).
10. – A light party game where players swap recipes in taverns, bluffing about their ingredients.

🎮

11. – City-builder where you oversee a Renaissance court laboratory, balancing resource extraction with courtly intrigue.
12. – A sim where you manage patronage, apprentices, and rivalries, while improving glass, porcelain, and dyes.
13. – A management game where artisanal workshops integrate alchemical methods to outperform competitors.
14. – Turn-based strategy about networking between patrons, scholars, and artisans, with reputations rising and falling.
15. – A crime sim where you secretly run forgeries of gold, silver, and gems while maintaining a respectable workshop.
16. – Rival alchemists vie

for noble favor by producing spectacles (fireworks, sugar sculptures, automata).

17. – Multi-generational sim where you pass down recipe books, refining craft knowledge across centuries.

18. – Role-playing sim of balancing research, teaching, spectacle, and secrecy under a suspicious patron.

19. – A research sim where every material (gold, copper, crystal, dyes) has properties to test, manipulate, and exploit.

20. – A proto-capitalist sim where your innovations in pigments, glass, and porcelain feed into global trade routes.

🕹️

21. – Players are itinerant alchemists navigating city guilds, courts, and underground markets to survive.
22. – Political RPG where your every recipe could be proof of genius—or heresy.
23. – Character-driven RPG about apprentices learning artisanal and alchemical arts, risking betrayal.
24. – Heist-style RPG where players collaborate on scams involving false gems, gilding, and dyes.
25. – A sandbox RPG where you travel between courts, monasteries, and workshops gathering recipes.
26. – A mystery RPG uncovering rival artisans who sabotage experiments to secure monopoly.
27. – Hybrid

trade/adventure RPG where you manage caravans of dyes and minerals while secretly pursuing chrysopoeia.

28. – Narrative RPG about the human cost of porcelain research, with characters tied to courts and secrecy.

29. – Alchemy-noir RPG where lab explosions hide rivalries, and your contacts in taverns hold the key to recipes.

30. – Occult-tinged RPG where players combine alchemical philosophy with real trades (smithing, apothecary).

🎮

31. – An action-adventure where you sneak into rival workshops to steal recipes and rare minerals.
32. – A combat sim where improvised alchemical weapons (firepots, acids) are your arsenal.
33. – An action game about rival courts stealing each other's crystal/glass formulas.
34. – A stealth game where you infiltrate guilds and courts, planting counterfeit coins.
35. – An adventure game following a caravan trader balancing honest craft and alchemical secrets.
36. – Pyrotechnic platformer about smuggling saltpeter and making fireworks for festivals.
37. – Roguelike where you experiment with combinations of

substances under time pressure.

38. – A survival game where artisanal processes (smelting, glassblowing, distilling) are reimagined as mechanics.

39. – An adventure where your rise depends on presenting the right material marvels to patrons.

40. – A dark adventure where failed alchemists turn to espionage and sabotage in rival guilds' laboratories.

🔭 These try to capture the that defined the early modern world. Some lean into , others into —but all circle around the tension between .

Want me to (like sim/strategy or RPG) and sketch out a

for one of these?

USER

I was thinking a roguelike inspired by FTL, except instead of a ship you are building a lab, instead of a map of space you are looking at early modern europe, medieval islamicate civ, or the late antique roman empire. And instead of combat we have encounters representing the social and business life of the alchemist, starting with an apprenticeship, with potential legal consequences if the alchemist gets caught doing anything unethical. Give me 40 features for this vide game

ASSISTANT

Perfect — a in the mold of *FTL*, but steeped in early modern alchemy and artisanal culture.

Here’s a list of that could give your game depth, variety, and replay value:

🏛

1. – You travel city-to-city, each node offering unique guilds, courts, and dangers.
2. – Begin with a master, limited recipes, and meager tools.
3. – Alembic bench, furnace, library, apothecary shelves, each upgradable.
4. – Gain access to noble courts for funding, prestige, and dangerous demands.
5. – Retire/die, pass recipes and apprentices down to continue the legacy.
6. – Guild fines, inquisitorial trials, or protection if you keep on good terms.
7. – Start in Florence, Baghdad, or Alexandria; materials and risks vary by setting.

8. – Survive to produce: Philosopher’s Stone, perfect porcelain, or gain noble title.

---

9. – Charcoal, wood, or coal affect efficiency and risks.
10. – Recipes take time; do you multitask or focus?
11. – Mismanaging heat or volatile mixtures can destroy lab modules.
12. – Mercury leaks, spoiled herbs, shattered crystal.
13. – Assign them to grind pigments, distill alcohol, or steal recipes.
14. – Hide contraband substances (arsenic, false gold) from inspectors.
15. – Some are incomplete or coded; decoding takes time.
16. – Tyrian purple, lapis lazuli, rock crystal, needing trade or smuggling.
17. – Manage slag,

fumes, poisons to avoid suspicion or accidents.

18. – Glassblowing, metalworking, dyeing; all require specialized modules.

---

19. – Surprise audits: produce dues or proof of honest work.
20. – Dazzle a duke with fireworks, sugar sculptures, or gilded armor.
21. – Rumors about new recipes, or rivals spreading slander.
22. – Chance to buy rare dyes, minerals, or forbidden texts.
23. – May duel you with rhetoric, attempt sabotage, or offer partnerships.
24. – Monks may sponsor medicinal alchemy, or accuse you of heresy.
25. – Legal trials with outcomes shaped by reputation and bribes.
26. – “Cure my gout,” “Imitate Chinese porcelain,” “Turn lead to gold.”
27. – Shortcut recipes to impress patrons, but risk exposure.

28. – Poorly treated assistants might betray or steal.

---

29. – Sell pigments, medicines, glassware, or fireworks.
30. – Mint false coins, fake gems; profitable but high risk.
31. – Huge opportunities for sales or scandal.
32. – Ship pigments to Venice, saffron to Cairo, porcelain to Paris.
33. – Wars or plague disrupt access to metals, dyes, or herbs.
34. – Pay smugglers to move saltpeter or manuscripts past customs.

⚖️

35. – Balance between safe artisanal goods and risky alchemical promises.
36. – If your remedies fail, patients may die, lowering reputation.
37. – Sell poisons to assassins, or refuse and lose income.
38. – If caught counterfeiting or practicing forbidden arts.
39. – Do you keep secrets, or publish recipes for prestige but lose monopoly?
40. – Be remembered as fraud, innovator, martyr, or legendary master of nature.

🔭 This structure makes the game a , where instead of fighting pirates you're dodging inquisitors, rival guilds, and scheming patrons — and

instead of upgrading weapons you're expanding furnaces, alembics, and recipe books.

Would you like me to (with choices and consequences) so you can see how a single event would play out in this framework?

USER

give me 40 thoughts on how we could design the lab building section with attention to narrative design considerations, visual legibility of the interface, and applying other relevant game design disciplines (name and explain what you're applying as you go)

ASSISTANT

Great — let's zoom in on the of your roguelike, and think about it systematically. I'll give you , weaving in *narrative design*, *visual design &*

legibility, and *game design disciplines* like

I'll name the discipline as I apply it and explain why it matters.

🏛

1. – The furnace, alembic, or library should carry a snippet of history (e.g. “This furnace was modeled after Biringuccio’s workshop”), anchoring mechanics in lore.
2. – Call an upgraded furnace “Athanor of Hermes” instead of “Furnace II.” This ties progress to alchemical mythos.
3. – Treat lab parts as personalities: the noisy furnace, the temperamental alembic, the dusty library. They can “complain” or “shine” narratively.
4. – Burn marks, cracked glass, or cluttered shelves show narrative consequences of risky experiments.
5. – Give players the option to conceal

fraudulent tools in the lab, tying visual space to moral choices.

6. – Apprentices comment on lab upgrades (“This new still… perhaps it’s strong enough for aqua vitae?”).

7. – Some modules (like fireworks benches) double as narrative “set pieces” to wow patrons.

8. – Layout decisions reflect character identity: a “chemist-doctor” builds apothecaries, a “counterfeiter” hides coin molds.

9. – Unlocking recipes through modules tells a story of knowledge transmission (bookshelves full, annotated manuscripts).

10. – A fire in the furnace tells a different story if your lab is cramped versus well-spaced.

🎨

11. – Furnaces = red, liquids = blue, texts = brown. Helps players scan quickly.
12. – Vertical = distillation, horizontal = grinding, circular = mixing. Shapes map to meaning.
13. – Make each module readable at thumbnail size, like FTL rooms.
14. – Steam rising = active distillation; flickering embers = active furnace.
15. – Show a glowing outline when a module is overheating instead of hiding warnings in text.
16. – Shelves visibly fill with jars, bags, metals, instead of abstract counters.
17.

– High zoom = full lab overview; close zoom = inspect individual alembic or text.

18. – Use historically-inspired icons (sun for gold, moon for silver) to reduce cognitive load.

19. – Messier labs = lower efficiency, reinforcing order vs chaos visually.

20. – Dim labs with candlelight for secrecy; bright open-air workshops for honesty.

---

21. – Like FTL, placing modules near each other reduces time cost (e.g. furnace next to alembic = faster distillation).
22. – Putting volatile modules next to each other saves space but risks chain explosions.
23. – Assign apprentices shorter walking distances for efficiency.
24. – Library + furnace = unlocks texts on metallurgy; storage + apothecary = better medicine recipes.
25. – Modules degrade with use, creating resource sinks and tension.
26. – Labs can lean into pharmacy, metallurgy, or spectacle; modules reflect

that specialization.

27. – Forces hard choices: more storage or another furnace?

28. – Mercury spills trigger poison events; apprentices can fall ill if ventilation is poor.

29. – A false wall allows illicit coin molds, creating stealth mechanics.

30. – If multiple apprentices work simultaneously, pathfinding creates emergent chaos.

🧠

31. – Modules evolve visibly at each upgrade stage, rewarding player investment.
32. – A mundane bookshelf might unlock a secret recipe for fireworks.
33. – A poorly placed furnace prompts NPC commentary: “Master, the fumes choke the apothecary!”
34. – Quick upgrades (extra shelf) vs slow prestige (build observatory).
35. – The lab feels like *your* creation, not generic, because of unique clutter and layout.
36. – Paint walls, decorate shelves,

symbolic crests from patrons.

37. – Crackling fires, bubbling liquids, scribbling quills reinforce activity at a glance.

38. – The lab ledger doubles as your “stats screen,” keeping UI inside the fiction.

39. – Explosions or messy failures are humorous/dramatic, softening frustration.

40. – The lab layout itself becomes a narrative artifact: screenshots tell the story of your playthrough.

👉 Together, these keep the , applying principles from .

Would you like me to (like an ASCII/Tiles sketch

inspired by *FTL*) so you can visualize how modules and apprentices move around?

USER

how can we start simple so as not to overwhelm the beginning player, but have things like tech trees or other level ups and upgradables gradually reveal themselves, and give players reasons to take an interest in these further resources?

ASSISTANT

That's a key question — roguelikes live or die by , while still offering over repeated runs. Here's a breakdown of how you could structure this for your alchemy-lab roguelike, with some principles from folded in:

🎯

- – At the very start, the player has only *one active module* (a simple furnace or workbench). All early recipes (like grinding herbs or boiling salt water) run through it.

- – Instead of tracking silver, patronage, reputation, apprentices, and ingredients all at once, introduce just *one resource* (say, copper coins) and one type of trade (selling pigments).

- – Frame this simplicity diegetically: you're an *apprentice* in a master's cramped lab. It makes sense that you don't have options yet.

- – First few events are about negotiating rent, fetching ingredients, or pleasing your patron with something trivial. No dangerous chemistry yet.

🌱

- – As you earn money or prestige, new modules appear (library, distillation table, storage room). Each unlock adds a new system layer, but only after the player has mastered the previous one.

- – Instead of giving the player a huge recipe book, start with a handful and “discover” others via manuscripts, patrons, or gossip. This encourages curiosity.

- – First time you unlock a new system, tie it to a story encounter (e.g. a noble asks for rosewater → you’re guided into distillation).

- – Some mechanics are “hidden” behind others (e.g. smelting unlocks alloys, alloys unlock counterfeit coins).

🔄

- – Upgrading from furnace I → II visibly changes the art, sound, and outputs. The “toy” is more fun to use, not just numerically better.

- – Patrons escalate requests: pigments are fine early on, but nobles want fireworks, porcelain, or medical cures. These *require* higher tech.

- – The player feels *limited* by their current lab: without a library, they can’t decode certain texts; without storage, goods spoil. This nudges them to expand.

- – Even if a player “loses” a run, certain recipe fragments or

apprentices can be carried over to the next playthrough, making late-game systems worth chasing.

🕹️

- : Player only sees the basics: pigments, coins, and a tiny workshop.
- : Unlocks new “eras” of complexity: metallurgy, dyeing, basic pharmacy.
- : Advanced systems like counterfeiting, porcelain, fireworks, or large-scale patronage politics emerge.
- – A toggle for players who *want* the full system dump early (for experts).

📖

- – Apprentices gossip about secrets you don’t yet have access to, planting curiosity.
- – A “locked” lab room visible but grayed out keeps players intrigued (“What goes there?”).
- – Patrons name-drop real-world marvels (Meissen porcelain, Venetian cristallo, Tyrian purple), hinting that you *could* produce them someday.
- – Patrons won’t take you seriously until you demonstrate mastery of higher systems. Narrative ambition ties to tech ambition.

✨ In short:

1. (one room, one resource, a couple recipes).
2. with clear narrative reasons.
3. (prestige, patronage, more powerful outputs).
4. , keeping early plays approachable but later plays rich.

Do you want me to sketch a (what systems and modules unlock when, what story beats nudge them) so you can see how the complexity layers build?

USER

Going back to our lists of materials and processes I want you to think about the economics of early modern europe, how profitable these various lines of work might be for our alchemists, and how we can design interesting choices into the game for navigating these artisanal worlds

ASSISTANT

Got it 👉 — let's connect the we listed earlier with the , and then map that onto for your roguelike. This means looking at:

1. – some trades are stable, some speculative, some fraudulent.
2. – some were protected guild monopolies, others shady/illegal.
3. – scarcity, reputation, patronage, and risk vs. reward.

💰

- – Common, modest profits.
- – Everyday utility, steady demand, but guild-controlled.
- – Reliable but labor-intensive.

👉 *Game design role:* “safe” options for early play. Low income but low risk. Players might lean on these when reputation is shaky.

- – Lucrative in Venice but high skill and startup cost.
- – Profitable, but failure = lawsuits, or worse, inquisitors.
- – Big demand, controlled by merchants; margins fluctuate with imports.
- – Strong urban demand, especially among elites.

👉 *Game design role:* The “middle path” of play. Steady money, but prone to or .

- – Immediate huge profits, but extreme risk of execution.
- – Courts pay big, but failure can ruin reputation instantly.
- – Enormous prestige if successful, but research-heavy and prone to financial ruin.
- – Hugely profitable, but dangerous (mercury, arsenic).
- – If a patron believes you, you're rich; if they doubt you, you're ruined.

👉 *Game design role:* Temptations. These create dramatic, story-driving choices.

- – Courtly display, profitable only at festivals.
- – Luxury niche, good in specific cities/patrons.
- – High prestige, tied to long-distance trade.
- – Good profits, but can overlap with fraud detection.

👉 *Game design role:* “Situational

opportunities” that keep playthroughs varied.

⚖️

- Do you with steady dye sales, or chase a that could double your prestige or bankrupt you?

- Do you for quick cash, knowing inspectors may raid your lab at any time?

- Wars or plagues cut off indigo imports → players must adapt to local pigments.

- Certain cities specialize (Venice for glass, Florence for porcelain, Seville for cochineal) → route choice becomes economic strategy.

- High reputation unlocks with huge payouts (fireworks for a royal wedding).

- But courtly life also exposes you to .

- Low reputation pushes you toward shady trades (counterfeiting, poisons).

- Space in the lab is limited: invest in a furnace (metals) or an apothecary bench (medicine)?

- Money is finite: do you save up to expand storage (long-term gains), or gamble on exotic ingredients for a single commission?

- Fraud is profitable short-term, but if caught, your legacy might be "charlatan."

- Honest artisanal labor earns slow prestige, but builds toward

.

🕹️

1. – Indigo doubles in price after a naval blockade; arsenic is cheap but suspicious.
2. – Patrons request rare goods (porcelain, Tyrian purple) that may not be immediately possible.
3. – Every line of work has a “risk meter” (fraud suspicion, guild anger, inquisitorial attention).
4. – Your build choices define you as a “glassmaker,” “pharmacist,” or “forger.”
5. – Some goods (fireworks, porcelain) yield prestige more than money; patrons care about prestige.
6. – Option to “cut” expensive ingredients (use saffron substitute, fake gems), raising profit but increasing risk of exposure.

7. – You can buy expensive licenses to reduce legal risk in trades like dyeing or apothecary, trading money for safety.
8. – Research-heavy projects (porcelain, Philosopher's Stone) burn resources until breakthrough or ruin.
9. – Certain goods sell better at festivals (sugar sculptures at Carnival, fireworks at coronations).
10. – Rivals corner a market (like ultramarine) and you must undercut or sabotage them.

👇 , the economy becomes a .
The player's story emerges from which trades they pursue: the cautious pigment merchant, the daring counterfeiter, the noble glassmaker, the doomed porcelain dreamer.

Would you like me to
(rows = materials/processes,

columns = profitability, risk, prestige value, narrative hooks), so you have a design reference you can plug directly into your game system?

USER

give me designs for encounter templates that cover scenes in the social life of an alchemist including court patronage mapped out across europe, all the merchants and tradesmen he employs building the lab, and artists or rulers who benefit from his chemical productions

ASSISTANT

Perfect — this is where your roguelike can really sing: not just the , but the that give every run narrative tension. Think of these as (like *FTL* events) with multiple-choice outcomes, mixing history, economics, and roleplay. I'll group them

into three clusters: , , and .

👑

These represent the high-stakes world of

.

1.

- A Medici duke demands fireworks for a feast. Do you accept with cheap powder (risk of failure) or invest heavily for glory?

- Outcomes: Prestige gain, resource drain, or public humiliation.

2.

- The Emperor’s court seeks a cure for gout. Provide a risky experimental medicine, a safe herbal tonic, or stall for time.

- Outcomes: Enormous patronage, suspicion of fraud, or dismissal.

3.

- An Ottoman courtier asks for Venetian-style cristallo glass. Do you risk smuggling techniques across borders?

- Outcomes: Riches, trade ban, or espionage accusation.

4.

- At Rudolf II's court, you're asked to prepare strange materials for astrological experiments. Cooperate, mock, or secretly substitute.

- Outcomes: Prestige, occult reputation, or exposure.

5.

- A papal official inspects your lab for heretical practices. Bribe, justify through natural philosophy, or hide contraband.

- Outcomes: Reputation boost, fine, or Inquisition risk.

6.

- Tsar demands porcelain like China's. Attempt research, bluff with fake glaze, or decline.

- Outcomes: Court favor, ruinous costs, or imprisonment.

🛠️

These represent the
that sustains the lab.

7.

- Fuel is low; a coal trader offers poor-quality stock cheap. Accept or pay premium for safety?

- Outcomes: Savings, risk of furnace explosion, or stable production.

8.

- Guild demands fees for using furnaces. Pay dues, negotiate, or smuggle glass secretly.

- Outcomes: Stability, reputation hit, or black-market access.

9.

- A shady trader offers mercury or cochineal illegally. Buy or decline?

- Outcomes: Rare recipes unlocked, guild suspicion, or arrest.

10.

- Your apprentice demands higher pay or threatens to leave. Raise wages, offer teaching, or dismiss him.

- Outcomes: Loyalty, resentment, or labor shortage.

11.

- A carpenter offers to expand your lab cheaply. Accept quick construction, pay

premium for safety, or wait.

- Outcomes: Expansion, fire risk, or delay.

12.

- A rare alchemical text is for sale. Pay dearly, try to copy, or dismiss as a fake.

- Outcomes: Unlock recipes, waste resources, or miss opportunity.

13.

- Local apothecaries accuse you of stealing their customers. Bribe, offer collaboration, or challenge them publicly.

- Outcomes: Market stability, new recipe exchange, or reputation risk.

14.

- Supply of copper or tin is cut. Pay bribes to miners, wait it out, or seek alternative materials.

- Outcomes: Delays, resource drain, or innovation.

15.

- A farmer offers crude saltpeter scraped from dung heaps. Refine it yourself, pay for purity, or refuse.

- Outcomes: Save money, risky failures, or delayed projects.

🎨

These are about

.

16.

- A painter begs for ultramarine, but you only have azurite. Sell him the cheap version, invest in costly import, or experiment with a fake.

- Outcomes: Money now, prestige later, or fraud risk.

17.

- Clerics need pigments for stained glass windows. Provide stable colors or experiment with volatile arsenic hues.

- Outcomes: Reputation, steady profit, or accident.

18.

- A theater troupe wants fireworks for a festival. Cheap powder, expensive rockets, or decline?

- Outcomes: Profit, prestige, or embarrassment.

19.

- A goldsmith asks for improved alloys. Deliver reliable bronze, attempt exotic mix, or hoard for yourself.

- Outcomes: Collaboration, innovation, or rivalry.

20.

- A king demands a fountain that spews wine and fire. Use honest hydraulics, alchemical trickery, or stall.

- Outcomes: Huge prestige, ruinous failure, or political fallout.

21.

- Court musicians want an alchemical automaton to play instruments. Accept, adapt recipes, or refuse.

- Outcomes: Prestige, wasted resources, or ridicule.

22.

- A textile merchant wants brilliant scarlet. Sell cochineal, try madder, or fake it.

- Outcomes: Trade profit, satisfied merchant, or suspicion.

23.

- Commission for massive sugar sculptures. Invest heavily, cut corners, or decline.

- Outcomes: Court favor, financial ruin, or scandal.

24.

- A scholar challenges your work. Debate openly, duel with experiments, or walk away.

- Outcomes: Prestige, humiliation, or neutrality.

25.

- A noble hints at needing a poison. Provide, refuse, or fake an antidote.

- Outcomes: Wealth, moral corruption, or danger if discovered.

🕹️

- → Florence, Prague, Istanbul, Venice, Rome, Paris. Each city has weighted encounter tables: Venice = glass, Florence = dyes/Medici, Prague = occult/Rudolf II, etc.

- → “Do you give the duke cheap powder?” Each option should tie into mechanics: *risk meters*, *prestige economy*, *money vs. reputation*.

- → Encounters affect lab (new recipes, fires, fines), economy (resources up/down), or narrative arcs (patron loyalty, guild rivalry, legal suspicion).

Would you like me to (with choice branches, mechanical effects, and narrative text) so you can see exactly how they'd play out in your roguelike?

USER

create a list of encounters for an apprentice to go through, leading up to an important trip to a nearby city

ASSISTANT

Here's a , designed to feel like a first act of your roguelike run. They teach basic systems (lab work, social navigation, economy) while building anticipation for a (maybe Florence, Venice, or Prague depending on setting). Each one has narrative flavor, choice structure, and light consequences.

- The master leaves you in charge of tending a small furnace.

-

- Stoke carefully → safe success, learn furnace basics.
- Overheat to impress → risk fire, higher payoff if lucky.
- Slack off → furnace cools, master scolds you.

- You’re tasked with grinding pigments for a painter.

-

- Grind carefully → low profit, builds reliability.
- Experiment with substitutes → higher reward, risk poor quality.
- Pocket some pigment for yourself → seed early dishonesty.

- Another apprentice in town mocks your lack of skill.
-

- Duel with words/knowledge → test lore skill.
- Challenge in a small experiment → risk small explosion.
- Ignore → no gain, avoid conflict.

- A coal merchant offers you fuel, but your purse is light.

-

- Buy cheap, dirty coal → risk furnace inefficiency.
- Save for quality coal → slower progress but safer.
- Try to haggle → test social ability, possible savings.

- Your master sends you to buy herbs at market.

-

- Buy proper herbs → stable recipe success.
- Cut corners with weeds → pocket money, risk failed medicines.

- Chat with merchant → unlock rumor about nearby city market.

- You find a torn page from an alchemical recipe in the lab.

-

- Study alone → partial knowledge, chance of insight.
- Ask the master → builds trust, but you reveal your curiosity.
- Hide it → keep for yourself, leads to secret project later.

- A local guild asks for fireworks for a festival.

-

- Help cautiously → safe small prestige.
- Mix bold formulas → big prestige, big risk of failure.
- Refuse → safety, but rivals gain.

- A minor noble stops by, curious about your lab.

-

- Show honest work (pigments, simple cures).
- Bluff about advanced secrets → risk exposure.
- Offer free gift → minor favor earned.

- In the market you overhear news: caravans, shortages, rumors of other alchemists.

-

- Listen quietly → unlock future trade info.
- Spread rumor → sabotage rival, but risk exposure.
- Approach merchants → chance at better deals.

(Major Turning Point)

- Your master announces a trip to a nearby city for patronage or materials, and you are invited (or ordered) to accompany him.

-

- Prepare carefully (save resources, pack safe recipes).
- Bring contraband (poisons, counterfeiting tools) for secret profit.
- Focus on study (bring books, scrolls, to expand knowledge).

- This begins the , with new encounter tables in the city.

⚖️

- Each encounter has clear (teaching systems) and (introducing alchemy's dangers).

- Together they tell the story of an in a small town → leading naturally to the where the roguelike really opens up.

- They also train players to weigh , which is the heart of your game.

Would you like me to (with text, choices, and outcomes) so you can see exactly how it plays at the player's level?